

Chapter 3: Titrimetry

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Contents

- **Introduction**
 - Definition of Terminologies
 - Determination of equivalent point
 - Titration Set Up: Equipment for Titration
 - Step-by-Step Process of Titration
 - Steps to Do in Titration Problems
 - Calculations in Titrimetry
 - Requirements for a Successful Titration
 - Titration Curves
 - Standardized Solution
- **Types of Titration**
 - **Acid-Base titrations**
 - **Precipitometric titrations**
 - **Complexometric titrations**
 - **Redox titrations**
 - **Diazotization titration**

Learning Outcomes

At the end of the Lesson, students will be able to:

- Define titrimetry
- Discuss the principles of the different titrimetric methods
- Explain the different applications of titrimetric analysis
- Differentiate the types of Titrations
- Classify acid-base titration curves
- Operate titrimetry using their basic skills of titration

Introduction: Definition of Terminologies

- **Titrimetry**: refers to a group of methods of **quantitative analysis** in which an **analyte** is determined basing on its **stoichiometric reaction** with a reagent of established concentration introduced to a sample gradually until the analyte is consumed quantitatively.
 - Titrimetric methods are still **widely used in pharmaceutical analysis** because of their **robustness, cheapness and capability for high precision**.
 - The only requirement of an analytical method that they **lack is specificity**

- **Titration:** is an analytical method where a scientist can determine the unknown conc. or quantity of an analyte via the volumetric addition of titrant solution with a known conc.
- Used to determine the amount or conc. of a substance
- It is a quantitative measurement of an analyte in solution by
- it's complete reaction without a reagent.

Introduction: Definition of Terminologies

- **A titration** is specified as" a process of **determining the conc. of a sample (analyte)** by **adding known increments of substance (titrant)** with which it responds till exact chemical equivalence is accomplished
- In pharmaceutical analysis, titration is **mainly used for quantitative analysis (assay) of active pharmaceutical ingredients (APIs) and excipients** with the purpose of assessing the **purity of a given substance (analyte).**

- An analyte, component (in clinical chemistry), titrand (in titrations), or chemical species is a substance or chemical constituent that is of interest in an analytical procedure
- It is the substance being analyzed or measured, typically a solution of unknown conc.
- It is the substance that reacts with the titrant

Introduction: Definition of Terminologies...

- **Titrant:** is a solution of **known conc.** that is added (titrated) from the burette to another solution (analyte) to determine the conc. of a second chemical species.
 - The titrant may also be called the **titrator**, the **reagent**, or the **standard solution**
- **Titrate:** refer to the **process of adding a measured amount** of one solution to another solution until a desired endpoint is reached
 - The process is used to determine the concentration of a solution by reacting it with a known concentration of another solution.

Introduction: Definition of Terminologies...

- **Equivalence point** (**stoichiometric point**) of a chemical reaction:
 - Refers to the point at which chemically equivalent quantities of reactants have been mixed.
- **Endpoint** (related to, but not the same as the equivalence point):
 - Refers to the point at which the **indicator changes color** in a colorimetric titration
- The point at which the reaction or endpoint is achieved can be detected by a **change in color, pH, or other physical property**.
- **Titration curve**: is a graphical representation of the pH of a solution during a titration

Introduction: Determination of equivalent point

Methods to Determine the Equivalence Point

- The different methods to determine the equivalence point include:
- **pH indicator:**
 - It is a substance that changes **color in response to a chemical change**.
 - An acid-base indicator (e.g., phenolphthalein) changes color depending on the pH.
 - Redox indicators (e.g. **methylene blue**) are also frequently used
 - A drop of indicator solution is added to the titration at the start; when the color changes the endpoint has been reached, this is an approximation of the equivalence point.

Introduction: Determination of equivalent point

Methods to Determine the Equivalence Point...

- The different methods to determine the equivalence point include:
- **Conductance:**
 - The conductivity of a solution depends on the ions that are present in it.
 - During many titrations, the conductivity changes significantly.
 - For instance, during an acid-base titration, the H_3O^+ and OH^- ions react to form neutral H_2O .
 - This changes the conductivity of the solution.
 - The total conductance of the solution depends also on the other ions present in the solution (such as counter ions).
 - Not all ions contribute equally to the conductivity; this also depends on the **mobility of each ion** and on the **total concentration of ions** (ionic strength).
 - Thus, predicting the change in conductivity is harder than measuring it.

Introduction: Determination of equivalent point...

Methods to determine the equivalence point

- **Color change:**
 - In some reactions, the solution changes color without any added indicator.
 - This is often seen in **redox titrations**, for instance, when the different **oxidation states** of the product and reactant produce different colors.
- **Precipitation**
 - If the reaction forms a solid, then a precipitate will form during the titration.
 - A classic example is the reaction between **Ag⁺ and Cl⁻** to form the very **insoluble salt AgCl**.
 - Surprisingly, this usually makes it difficult to determine the endpoint precisely.
 - As a result, precipitation titrations often have to be done as back titrations.
- **Isothermal Titration Calorimeter**
 - Uses the **heat produced or consumed** by the reaction to determine the equivalence point.
 - This is important in biochemical titrations, such as the determination of how substrates bind to enzymes.

Introduction: Determination of equivalent point...

Methods to determine the equivalence point...

- **Thermometric Titrimetry**

- It is an extraordinarily versatile technique.
- This is differentiated from **calorimetric titrimetry** by the fact that the **heat of the reaction** (as indicated by temperature rise or fall) is not used to determine the **amount of analyte in the sample solution**.
- Instead, the equivalence point is determined by **the rate of temperature change**.
- Because thermometric titrimetry is a **relative technique**, it is **not necessary to conduct** the titration under **isothermal conditions**, and titrations can be conducted in **plastic or even glass vessels**, although these vessels are generally enclosed to prevent stray draughts from causing "noise" and disturbing the endpoint.
- Because thermometric titrations can be conducted under ambient conditions, they are especially **well-suited to routine process** and **quality control in industry**.
- Depending on whether the reaction between the **titrant & analyte** is **exothermic** or **endothermic**, the temperature will either **rise or fall during the titration**.
- When all analyte has been consumed by reaction with the titrant, a change in the rate of temperature **increase or decrease reveals** the equivalence point and an inflection in the temperature curve can be observed.
- The equivalence point can be located precisely by employing the second derivative of the temperature curve.

Introduction...

Principles of Titrations

- An analyte is chemically reacted with a standard solution of a reagent of precisely known concentration or with a concentration that can be precisely determined.
- The **amount of a standard solution** required to completely react **with all of the sample** is used to estimate the purity of the sample

Applications

- **Provide standard pharmacopoeial methods** for the assay of unformulated drugs and excipients and some formulated drugs, e.g. those that **lack a strong chromophore**.
- Used for **standardizations of raw materials** and intermediates **used in drug synthesis in industry**.
 - Suppliers of raw materials may provide these materials at a **specified purity** which has been assayed titrimetrically to a pharmacopoeial standard.
- Certain specialist titrations, such as the Karl Fischer titration used to **estimate water content**, are widely used in the pharmaceutical industry.

Introduction...

Advantages of Titrations

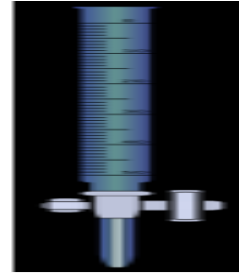
- Capable of a **higher degree of precision & accuracy** than instrumental methods of analysis, with precisions of $\alpha \pm 0.1\%$ being achievable.
- The methods are generally **robust**.
- Analyses can be **automated**.
- **Cheap to perform** and **do not require specialized apparatus**.
- They are absolute methods & are **not dependent on the calibration** of an instrument.
- It can be used to measure a **wide range of concentrations**
- Relatively **simple to set up**, which makes it ideal for students who are just starting to experiment with titrations in the laboratory

Limitations of Titrations

- **Non-selective**
- **Time-consuming** if **not automated** & require a greater level of operator skill than routine instrumental methods.
- **Require large amounts of sample and reagents**.
- Reactions of standard solutions with the analyte should be **rapid** and **complete**.
- The **accuracy of the results** can be **affected by external factors** such as the temp. of the reactants and the kind of indicator used
- **Precise measurements & equipment** are required to obtain accurate results

Titration Set Up: Equipment for Titration

- The setup for a titration can vary depending on the specific chemical reaction being studied but typically involves the following components:
 - **Analyte**
 - **Titrant**
 - **Burette**
 - **Indicator**
 - A substance that is added to the analyte to signal the endpoint of the reaction, usually by changing color or pH
 - **Flask or beaker:**
 - **Stirring apparatus:** A magnetic stirrer
- To set up a titration, the burette is first filled with the titrant solution and any air bubbles are removed.
- The analyte solution is then placed in a flask, and the indicator is added if necessary.
- The burette is carefully positioned over the flask, and the titrant is slowly added to the analyte while stirring continuously, until the endpoint is reached.
- The volume of titrant required to reach the endpoint can be used to calculate the conc. of the analyte.



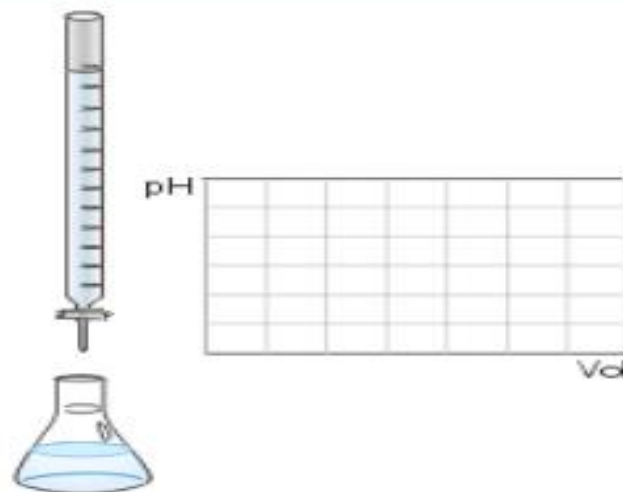
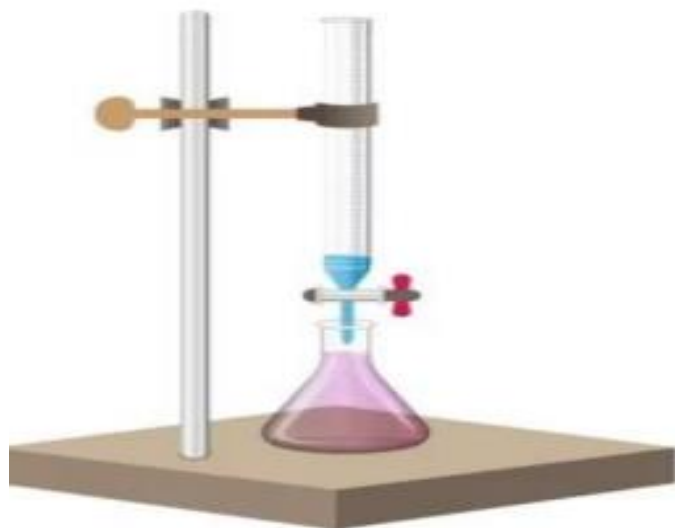
Step-by-Step Process of Titration

- Although there are many different types of titrations, all require the same basic steps for successful completion.
- Here is a step-by-step guide to the titration process:
- **Step 1: Prepare the Titrant**
 - Accurately measure & prepare the titrant.
- **Step 2: Prepare the Sample**
 - Prepare the sample containing the unknown analyte by weighing & diluting the sample, based on its composition & the type of titration being performed.
- **Step 3: Dispense the Titrant into the Sample**
 - Dispense the prepared titrant into the sample by slowly and accurately, using a burette or other dispensing device.
- **Step 4: Monitor the Reaction**
 - Monitor the reaction carefully.
 - Depending on the type of titration being performed, this may involve the use of a pH indicator (for acid/base titrations), a calorimeter (for redox titrations), or other means.
- **Step 5: Calculate the Concentration of the Analyte**
 - The final step in a titration is to **use the measured reaction to calculate the conc. of the unknown analyte.**
 - This is usually done by calculating the **molar concentration** of the **analyte** from the **concentration of the titrant** and the **volume added**

Steps to Do in Titration Problems

Titration problems can be solved by following the following steps:

- **Step 1: Calculate the Molarity (M) of the Titrant**
 - Calculate the molarity of the titrant (usually given on the label of the titrant solution)
- **Step 2: Calculate the Volume of Titrant Needed**
 - Calculate the volume of titrant that is needed to completely react with the amount of analyte present using the formula
- **Step 3: Choose an Indicator**
 - Choose an indicator to accurately identify the endpoint of the titration
 - The indicator should be selected so that it changes colour at the equivalence point.
- **Step 4: Perform the Titration**
 - Perform the titration by slowly adding the titrant to the analyte, using a burette, until the endpoint of the titration is reached.
 - ✓ The endpoint is the point where the indicator has changed color, indicating the titration is complete.
- **Step 5: Calculate the Concentration of the Unknown Solution**
 - Once the endpoint has been reached, the concentration of the unknown solution can be calculated using the formula:
 - **Concentration of Analyte (C) = Volume of Titrant (V) X Molarity of Titrant (M)**



Endpoint and Equivalence Point

- Though the terms equivalence point and endpoint are often used interchangeably, they are different terms.
- **Equivalence point:**
 - It is the **theoretical completion of the reaction:**
 - The **volume of added titrant** at which the **number of moles of titrant** is equal to the **number of moles of analyte**, or some **multiple thereof** (as in polyprotic acids).
- **Endpoint:**
 - It is what is **actually measured**, a **physical change** in the solution as determined by an **indicator** or an **instrument**
- There is a slight difference between the endpoint and the equivalence point of the titration.
 - This error is referred to as an indicator error & it is indeterminate

Calculations in Titrimetry

- The titrant is added either **manually** from a burette or **automatically** from an automatic titration apparatus (an automatic burette or titrator)
- Detection of the **end point** can be based on:
 - Visual inspection of colour change (Indicator detection) or
 - Electrochemical measurements (Potentiometric endpoint detection).
- In **Potentiometric endpoint detection**, the titration is termed **Potentiometric titration**, which is **preferred in the pharmaceutical industry** because it can be automated.
- The **volume of titrant consumed** is read from the **burette/titrator** and the **quantity of analyte** in the sample solution can be **calculated** on the basis of the underlying titration equation and the exact conc. of reagent in the titrant.

Calculations in Titrimetry...

- Titrations are based on **measurement of volumes**.
- An **accurate amount of analyte** is dissolved in a specific volume of solution, which is termed the **titrate**.
- Increments of a standardized solution of a reagent are added gradually until the entire amount of analyte has reacted.
 - The standardized solution is termed the **titrant**.
- The **conc. & exact volume of titrant added** are **used to calculate the amount of analyte**.
- More generally, a titration can be described by the following **titration equation**:
 - **$x(\text{Analyte, A}) + y(\text{Titrant, T}) \rightarrow \text{Products}$**
 - Here **x** and **y** are the number of mol of **analyte** (pharmaceutical compound) and **titrant involved** to complete the titration, respectively
- For volumetric titrations, the amount, n_A , of analyte in the sample can be calculated using

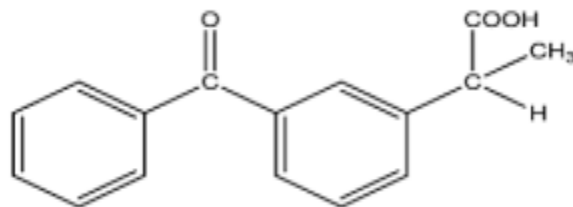
$$n_A = \frac{x}{y} C_T V_T$$

- The titration is completed when all the analyte has reacted & transformed to products.
 - The titration has then reached the **equivalence point**.

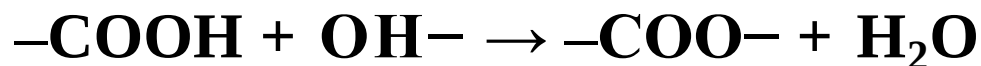
Calculations in Titrimetry...

Example: Titration of a pharmaceutical ingredient & calculation of purity (assay)

- To establish the purity of the **API** of **ketoprofen**, **0.1996 g** is weighed on an analytical balance and dissolved in 25 mL of ethanol 96% (v/v); 25 mL of water is added to the solution (note that knowledge of the exact volume of the final solution is not needed in this case as the exact mass of the analyte is known). The solution is titrated with **7.85 mL of 0.1006 M NaOH** to the equivalence point. The structure of ketoprofen is illustrated below:



- Ketoprofen is an acid and the carboxylic group (–COOH) will react with NaOH during the titration according to the follow scheme:



Calculations in Titrimetry...

Example: Titration of a pharmaceutical ingredient and calculation of purity (assay)...

- As seen from the reaction, ketoprofen and NaOH react in the molar ratio 1:1, and the number of mol of NaOH used during the titration is, therefore, equal to the number of mol of pure ketoprofen present in the 0.1996 g portion of API.

- The number of mol of NaOH (n_{NaOH}) reacted is calculated as follows:

$$\text{Molar Conc. (M)} = \frac{\text{Moles of Solute, } n_{\text{NaOH(mol)}}}{\text{Liters of Solution (L)}}$$

$$n_{\text{NaOH}} = C \times V = 0.1006 \text{ mol/L} \times 0.00785 \text{ L} = 7.89 \times 10^{-4} \text{ mol}$$

- This is equal to the number of mol of pure ketoprofen present in 0.1996 g portion of API:

- Mass of Ketoprofen = Number of Moles n x Mol. wt**

$$= 7.89 \times 10^{-4} \text{ mol} \times 254.3 \text{ g/mol} = 0.2006 \text{ g}$$

- The **purity of the active ingredient** is calculated as follows:

$$\text{Purity} = \frac{0.2006}{0.1996} \times 100\% = 100.5\%$$

Requirements for a Successful Titration

- Despite the difference in chemistry, all titrations share several common features
 - The titration reaction must be **well-defined** and **without any side reactions**
 - The **stoichiometry of the reaction** must be **known** in order to calculate the amount of analyte in the sample
 - The reaction must be virtually **complete** ($\approx 100\%$ of analyte must be converted to product).
 - Other substances in the sample should **not react with the titrant**
 - The **equivalence point** should be **clearly detected**
 - A **high reaction rate** is advantageous, but not essential, as a **fast** reaction between the analyte & the reagent reduces the analysis time
 - The reaction should be **quantitative** so that **sharp change will occur at the end-point**
 - The **exact concentration of titrant** must be known (standardized solution)

Titration Curves

- Titration curve is plot of a variable related to a relevant concentration as y-axis versus some measure of the amount of titrant (e.g. volume) on the X-axis.
- A titration curve provides us with a visual picture of how a property of the titration reaction changes as we add the titrant to the titrand.
- Titration curve **define the required properties of an indicator** and allow us **estimate the error** associated with the titration methods.

Titration Curves...

Example:

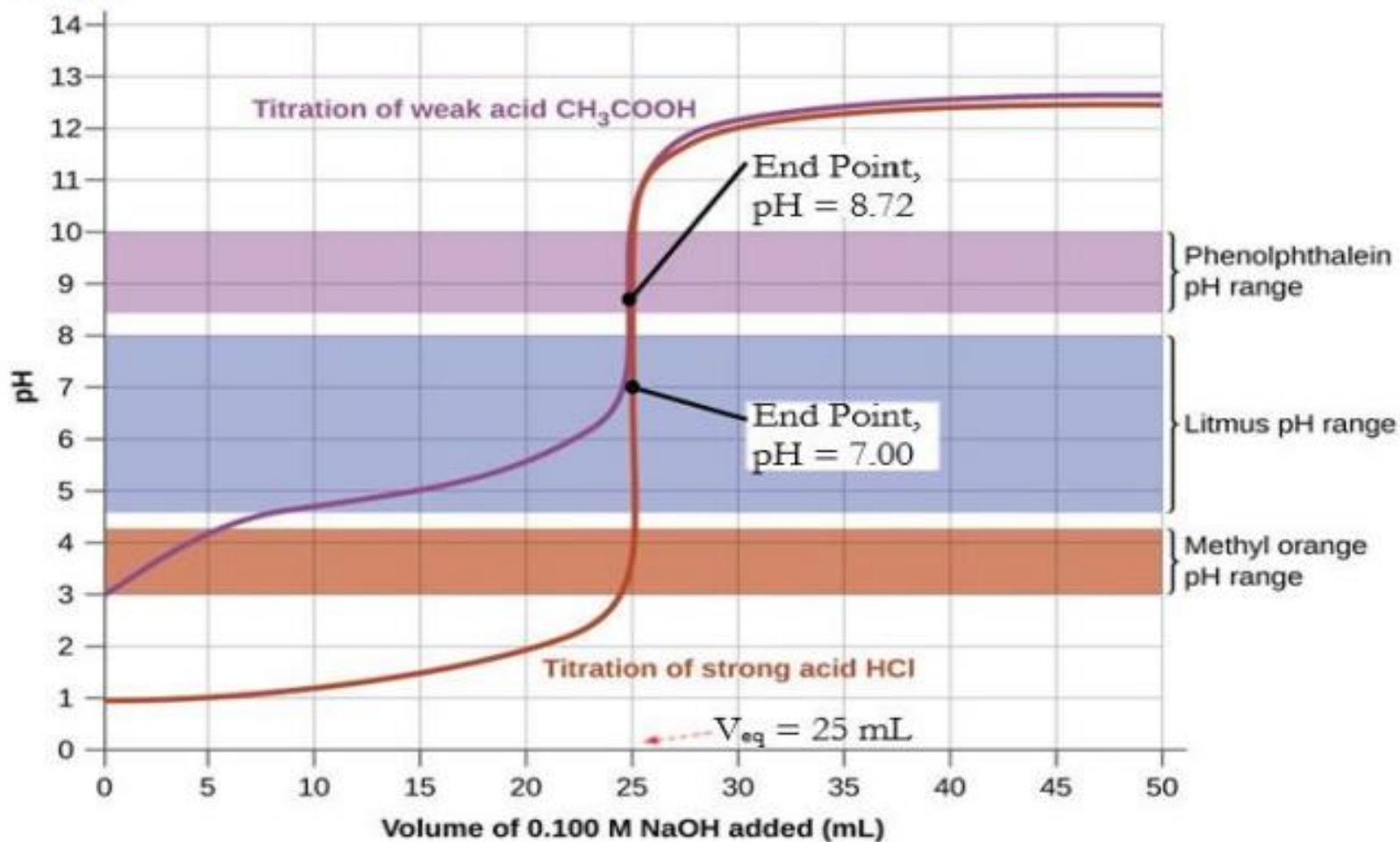


Fig: Acid-base titration curve

Standardized Solution

- During titration, the **titrant** is added to the **titrate** until the equivalent point is reached, where virtually all analyte has been converted into products.
- All titrations are equilibrium reactions, but those used in pharmaceutical analysis have equilibrium completely shifted to the product side.
- The titration is stopped when the **endpoint** of titration is reached
 - Ideally, the endpoint coincides with the equivalence point. If not, the result is prone to **titration error**.
- To **achieve high accuracy in titration**, it is essential to know the exact conc. of titrant.
 - Thus, the titrant has to be **standardized** prior to use.
- **Standardized Solution are of two types:**
 - **Primary Standards** and **Secondary Standards**
 - ✓ For standardization, a **primary standard** is needed.

Standardized Solution...

- A **primary standard** is a chemical reagent with the following characteristics:
 - **High chemical purity** (>99.95%) that serves as a reference
 - **Well-defined chemical composition** (including the amount of crystal water)
 - **High stability** during storage or in contact with air & light
 - **High solubility** in the titration solvent (water or organic solvent)
 - Non-volatile
 - Strong electrolyte

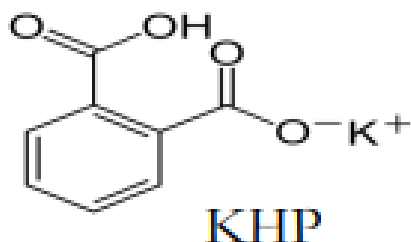
Table: Examples of Primary Standards in Ph. Eur.

Primary Standard	Potassium Hydrogen Phthalate (KHP)	Anhydrous Na_2CO_3	KBrO_3	Benzoic Acid
Mol. Wt.	204.2 g/mol	106 g/mol	167 g/mol	122 g/mol

Table: Primary Standards and their Uses

Primary standard	Uses
Potassium Hydrogen Phthalate	Standardization of NaOH solution
	Standardization of acetic perchloric acid
Potassium iodate (KIO_3)	Standardization of sodium thiosulphate solution through generation of iodine
Anhydrous Na_2CO_3	Standardization of hydrochloric acid
Zinc metal	Standardization of EDTA solution

EDTA: Ethylenediamine tetracetic acid



Standardized Solution...

- Standards that do not meet all the Primary Standards criteria are called **Secondary Standards**
 - Exact concentration is determined by standardization
- **Standard solutions are prepared in two ways:**
 - **Direct Method:**
 - Can be made directly by **accurate weighing of the primary standard** (e.g. KHP) and subsequent **dissolution & dilution** to a specific volume.
 - The concentration of such solutions is known with high accuracy.
 - **Indirect Method:**
 - Standardization will be performed (NaOH vs KHP)

Back Titration

- **Back titration:** is a titration done in reverse; instead of titrating the original sample, a **known excess of standard reagent is added to the solution**, and the **excess is titrated**.

When do we use back titration?

- If the **end point of the reverse titration is easier to identify** than the **end point of the normal titration** as with **precipitation reactions**.
- If the reaction between the analyte and the titrant is **very slow**.
- When the analyte is **Volatile substance**
 - E.g. Ammonia, some of which would be lost during the titration.
- When the analyte is in a **non-soluble solid (Insoluble substance)**
 - E.g.: Require excess volumetric reagent to effect a quantitative reaction.
- **Substance that requires heating** with a volumetric reagent during the determinations in which decomposition or loss of reactant or product would occur in the process.
 - E.g.: Formaldehyde solution

Types of Titration

Aqueous Titration

- There are two types of titration:
 - **Aqueous Titration and Non-Aqueous Titration**
- Aqueous titration refers to the process of titration that executes in an aqueous solution.
- Aqueous titration is the titration process which utilizes water in place of sample solvent for the determination of the concentration of solute in the sample.
- In Pharmaceutical Analysis, some **aqueous titrations** are used include:
 - **Aqueous Acid-base titration,**
 - **Aqueous Redox Titration,**
 - **Precipitation Titration, and**
 - **Complexometric Titration.**

Non-Aqueous Titration

- It is defined as the process in which solute liquefies in a non-water containing solvent.
- It is the most common titrimetric procedure used in **pharmacopoeial assays** and serves a double purpose,
 - As it is suitable for the titration of **very weak acids** and **bases** and
 - **Provides a solvent** in which **organic compounds are soluble**.
- The most commonly used procedure is the titration of **organic bases** with **perchloric acid in acetic acid**.

Non-Aqueous Titration...

Types of Non-Aqueous Titration and their Significance

- It is the titration of **weakly acidic or basic analytes** in a solvent medium that lacks water.
- The solute is dissolved in a non-aqueous solvent, and is further titrated with an **acid or base titrant**.
- The **principle of acid-base titration** and **non-aqueous titration is the same**.
 - It states that an acid donates a proton to another chemical and a base accepts a proton.
- The interference of water molecules in titration is the rationale for non-aqueous titrations.
 - Water is a weak acid and a weak base.
 - Water molecules compete with various bases and acids dissolved in them for proton donation and acceptance.
 - Water competition makes it harder to determine the titration's endpoint.
 - As a result, a titration process that does not include the interference of water molecules is required; hence, non-aqueous solvents are utilized to assess chemicals, a process known as non-aqueous titration.

Non-Aqueous Titration...

Types of Solvents Used in Non-Aqueous Titrations

- There are **4 types of solvents used in non-aqueous titration**.
- **Aprotic Solvents:**
 - In this form of non-aqueous solvent, a hydrogen atom is bound to oxygen, fluorine, or nitrogen.
 - These are chemically inert and have a neutral charge.
 - Examples are **chloroform** and **benzene**.
- **Protogenic Solvents:**
 - These are **acidic solvents** that **quickly donate protons**.
 - They are usually used to boost the alkalinity of weak acids and strengthen weak bases
 - Examples are **acetic** and **sulphuric acid**.
- **Protophilic Solvents:**
 - These are **basic solvents** with a **high proton acceptance tendency**.
 - They create solvated protons when they react with acids.
 - Examples are **amines** and **ammonia**.
- **Amphiprotic Solvents:**
 - These are solvents with **acidic and basic qualities**, as well as **protogenic & protophilic solvent properties**.
 - **Alcohols** and **acetic acid** are two examples.

Non-Aqueous Titration...

Indicators Used for Non-Aqueous Titrations

- The following indicators are used in non-aqueous titrations:
 - The **resonant, ionized & unionized forms** of indicators are used; however, the **colour change at the endpoint of reaction** varies among titrations depending on the **titrant nature**.
- **Crystal Violet Indicator:**
 - Widely used and employed in glacial acetic acid titration as a 0.5%w/v solution
 - It is considered as the most common indicators in the titration of the bases
 - It gives **violet colour** in **basic medium** and **yellowish green** in **acidic medium**
- **Methyl Red Indicator:**
 - It is made in a 0.2% w/v dioxin solution and its endpoint indicates a change in colour from **yellow to red**.
- **Naphtholbenzein Indicator:**
 - It is made in a 0.2% w/v solution as **ethanoic acid** and indicates a colour change from **yellow to green at the endpoint**.
- **Quinaldine Red:**
 - Most drug assays use it, and its colour shifts from **purplish red** to **light green at the endpoint**.
- **Thymol Blue:**
 - It is frequently used as a **pH indicator** in various titrations.
 - It is utilized as a 0.2% w/v solution in methanol, with a **yellow to blue** colour change.

Non-Aqueous Titration...

Advantages of Non-Aqueous Titration

- It is used to titrate **organic acids or bases** that are **insoluble in water** and **soluble in non-aqueous solvents**.
- It can titrate **weak acids or bases** that are **impossible to titrate in water**.
- **Organic acids** with a strength comparable to water can be easily titrated.
- It can also be used to titrate samples that contain **acid combinations**.
- It is **easy to carry out** and **does not require any specific laboratory equipment or glassware**.
- Internal indicators can deliver a quick and precise result.
- This titration is **crucial in pharmacopoeial assays** and for **biological substances**.
- The potentiometric approach can be used to **automate the operation**.

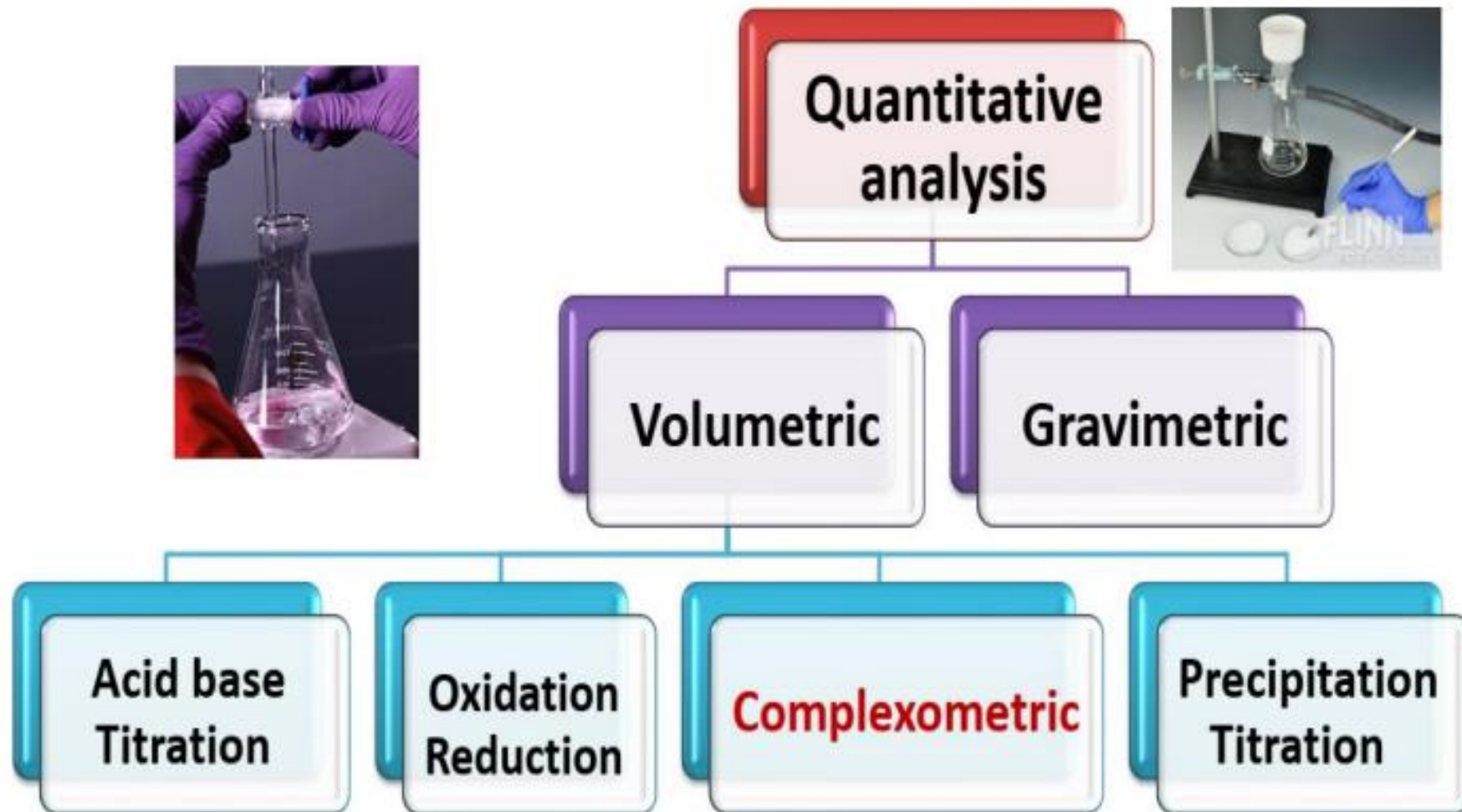
Non-Aqueous Titration...

Disadvantages of Non-Aqueous Titrations

- Because this is an open system, **temperature and moisture** may impact the outcomes; hence, they must be **controlled**.
- The non-aqueous titration solvent is **more expensive** than the aqueous titration solvent.
- It is **non-specific**, **contamination may occur**.
- A **volatile solvent** can **contaminate the environment** more than aqueous solvents.
- The non-aqueous solvents should be **calibrated after each usage** in non-aqueous titration.

Types of Aqueous Titration

The **type of reaction** provides a simple way to divide titrimetry into the following categories:



Acid-Base Titration

- It is a method of quantitative analysis for **determining the conc.** of an **acid or base** by **exactly neutralizing** it with a standard solution of **base or acid having known conc.**
- It is an experimental technique **used to acquire information** about a **solution containing an acid or base.**
 - In this type of titration, a **standard acid is titrated with a solution of a strong base**, or **vice versa**.
- An **indicator** is used to identify when the end point has been reached, which occurs when the acid and base react to form a salt and water
 - For example: **phenolphthalein** or **methyl orange**
- For an acid-base reaction the equivalence point is where the moles of acid and the moles of base would neutralize each other according to the chemical reaction.



Acid-Base Titration...

- **Two Types**
 - **Aqueous Acid-Base Titration**
 - **Non-aqueous Acid-Base Titration**
- **The Aqueous Titration:** Defined as the process of titration that executes in an aqueous solution.
- **Non-aqueous Titration:** Defined as the process in which solute liquefies in a non-water containing solvent.
- The **main difference** between aqueous and non-aqueous titration:
 - **Aqueous titration utilizes water as a solvent** and **non-aqueous titration utilizes organic solvent.**
- Other differences between aqueous and non-aqueous titration is their types:
 - **Types of Aqueous Titration**
 - Acid-base titration, aqueous redox titration, complexometric titration & precipitation titration
 - **Types of Non-aqueous Titration**
 - Acid-base titration, Non-aqueous redox titration, Iodometry and Iodimetry.
- **Different indicators are used in both titration processes:**
 - In aqueous titration **litmus (azolitim)** is used as an indicator and in non-aqueous titration **crystal violet** is used as an indicator.

Acid-Base Titration...

- **Common Titrants**

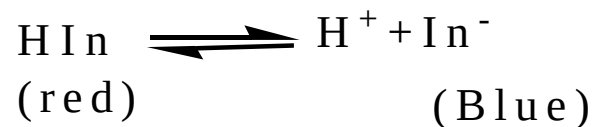
- In order for the titration reaction to go to completion, a **strong acid or a strong base** is the usual choice for a titrant in acid-base titrations.
 - For the analysis of bases: HCl , H_2SO_4 or HClO_4
 - For the analysis of acids: NaOH , KOH or $\text{Ba}(\text{OH})_2$

- **Detection of the end point: Indicators**

- It is usually more convenient to add an **indicator** to the solution and **visually detect a color change**.
- An indicator for acid-base is a **weak acid or weak base** that is highly colored.
- The color of the **ionized form** is markedly different from that of the **non-ionized form**

Acid-Base Titration...

- **Detection of the end point: Indicators**
 - Assume the indicator is **weak acid**, designated **HIn**, and assume that the non-ionized form is **red** while the ionized form is **blue**



- One can write a **Henderson-Hasselbalch equation** for this, just as other weak acids:

$$\text{pH} = \text{pK}_a + \log[\text{In}^-]$$

[HIn]

- With indicators in which **both forms are colored**, generally only one color is observed if the ratio of the concentration of the two forms is **10:1**
- When **only the color of the non-ionized form is seen**,
 $[\text{In}^-]/[\text{HIn}] = 1/10$

$$\text{pH} = \text{pK}_a + \log \frac{1}{10} = \text{pK}_a - 1$$

Acid-Base Titration...

- When only the color of the ionized form is observed,
 - $[\text{In}^-]/[\text{HIn}] = 10/1$

$$\text{pH} = \text{pK}_a + \log_{10} \frac{10}{1} = \text{pK}_a + 1$$

- So, the pH in going from one color to the other has changed from **pKa-1 to pKa+1**.
- Midway in the transition, the concentrations of the two forms are equal, and **pH = pKa**.
- Obviously, then the ***pKa of the indicator should be close to the pH of the equivalence point.***

Acid-Base Titration...

Classification: Based on the strength of the Acid and Base used (as titrant and Analyte)

- I. Direct titration of **Strong Acid with Strong Base**
- II. Direct titration of **Weak Acid with Strong Base**
- III. Direct titration of **Strong Base with Strong Acid**
- IV. direct titration of **Weak Bases with Standard Strong Acid**

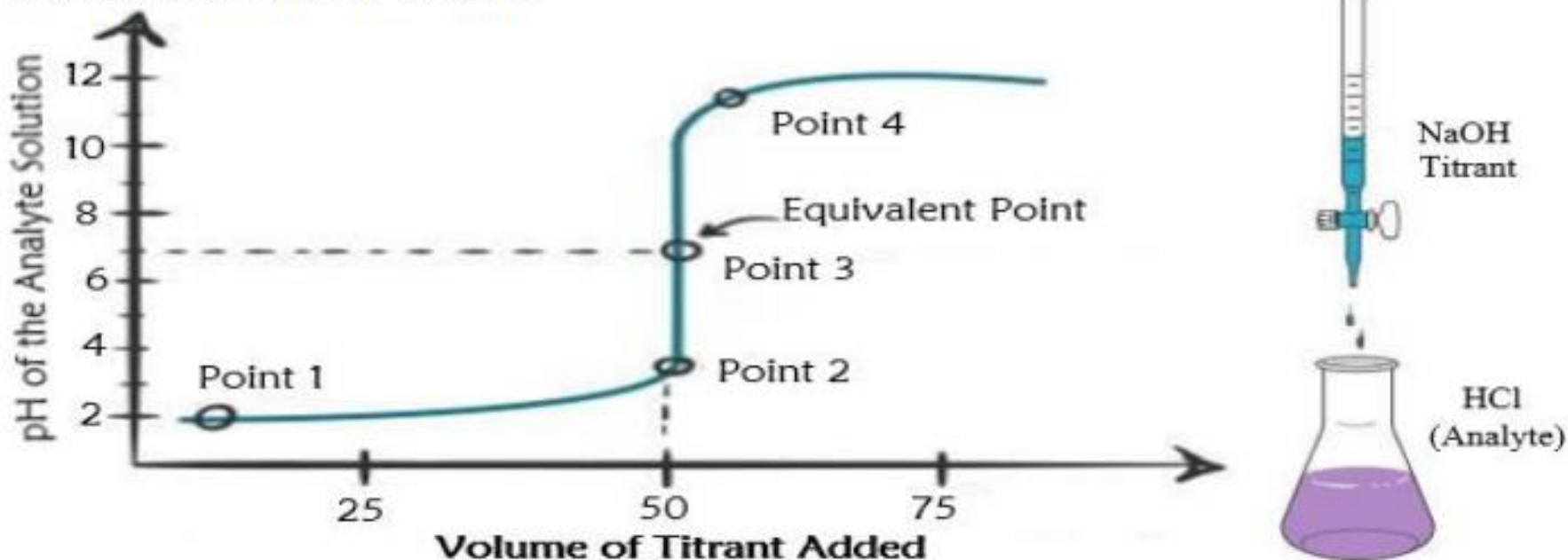
Acid-Base Titration...

- **I. Direct titration of Strong Acid with Strong Base**
 - Characterized with large pH change around the equivalence point; **Eq.pt at pH = 7**
 - **Before the equivalence point**, pH depends on amount of excess strong acid present
 - **After equivalence point**, pH depends on amount of excess base added.
 - Indicator which changes color in pH 4-10 is suitable (**Methyl orange**)

Acid-Base Titration...

- Example: Consider the titration of 50.0 mL of 0.100 M HCl using a titrant of 0.200 M NaOH. Construct the titration curve for this titration.
- The analyte is HCl (strong acid) and the titrant is NaOH (strong base).
-

If we start plotting the **pH of the analyte** against the **volume of NaOH** that it is added from the burette, we will get a titration curve as shown below:



Acid-Base Titration...

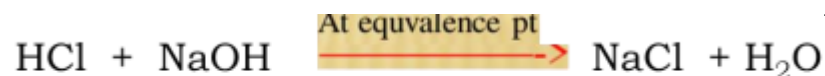
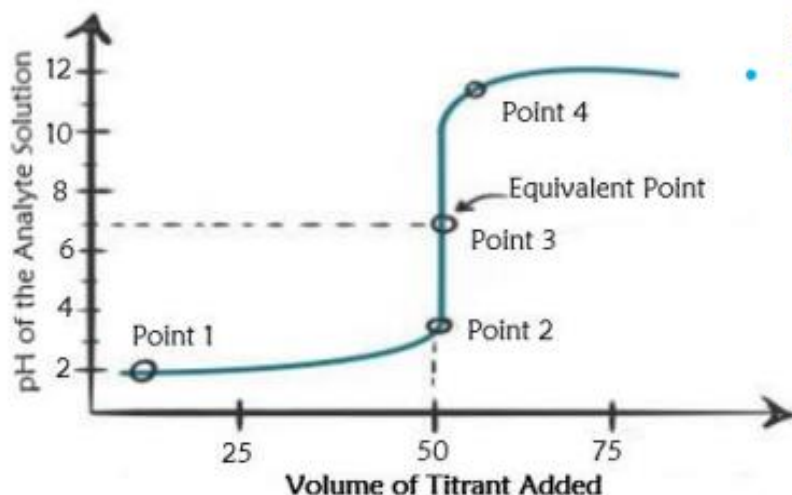
- **Point 1:** No NaOH added yet, so the pH of the analyte is low. As NaOH is added dropwise,

$$\text{H}_2\text{O} + \text{HCl} \rightarrow \text{H}_3\text{O}^+ + \text{Cl}^-$$
- As NaOH is added dropwise, H_3O^+ slowly starts getting consumed by OH^- produced by dissociation of NaOH, Analyte is still acidic due to predominance H_3O^+

- **Point 2:** This is the pH recorded at a time point just before complete neutralization takes place.

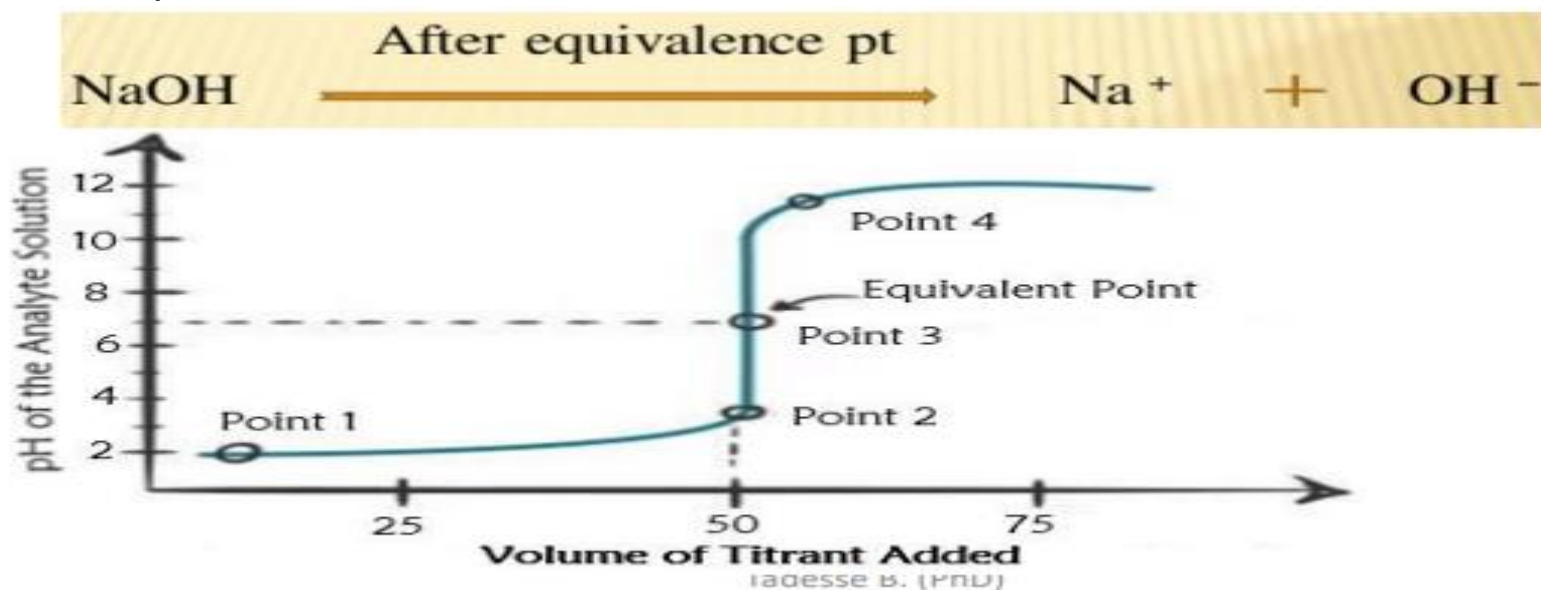
- **Point 3:** This is the equivalence point (halfway up the steep curve).

- At this point, **moles of NaOH added = moles of HCl in the analyte**
- At this point, H_3O^+ ions are completely neutralized by OH^- .
- The solution only has salt (NaCl) & water & therefore, the pH is neutral, i.e. pH = 7.



Acid-Base Titration...

- **Point 4:** Addition of NaOH continues, pH starts becoming basic because HCl has been completely neutralized and now excess of OH^- ions are present in the solution (from dissociation of NaOH)



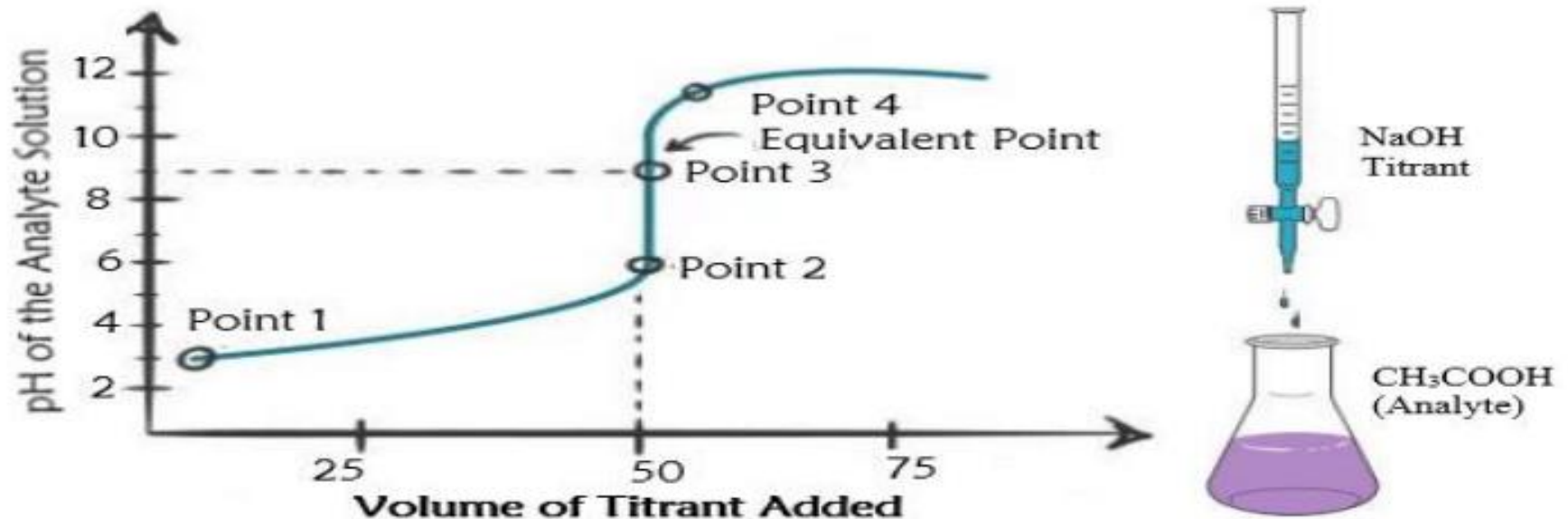
Acid-Base Titration...

- **II. Direct titration of weak acid with strong base**
- Equivalence point: $\text{pH} > 7$
- The transition range of the indicator for this titration of a weak acid must fall within a pH range of 7 to 10.
- Most frequently used indicator in such titration is
- **phenolphthalein.**
- There is small pH change around the equivalence point
- If indicator like Methyl red (4-6) is used, there will be change in the color before the equivalence point.

Acid-Base Titration...

II. Direct titration of weak acid with strong base...

- Assume the **analyte** is **acetic acid** (CH_3COOH , weak acid) and the **titrant** is **NaOH** (strong base). If we start plotting the **pH of the analyte** against the **volume of NaOH** that we are adding from the burette, we will get a titration curve as shown below:



Tadesse B. (PhD)

yohannes G.

51

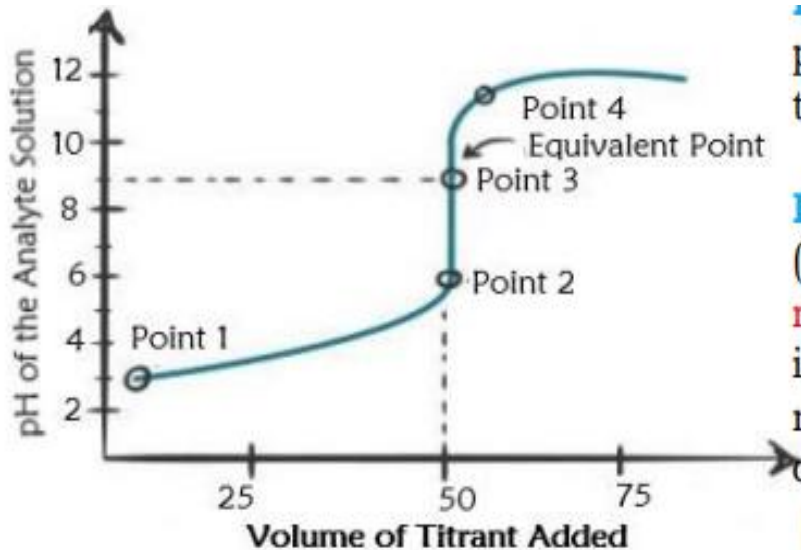
53

Acid-Base Titration...

- Point 1:** No NaOH added yet, so the pH of the analyte is low (it predominantly contains H_3O^+ from dissociation of CH_3COOH). But acetic acid is a weak acid, so the starting pH is higher.



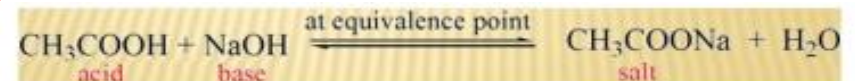
- As NaOH is added drop wise, H_3O^+ slowly starts getting consumed by OH^- (produced by dissociation of NaOH). But the analyte is still acidic due to predominance of H_3O^+ ions



Point 2: This is the pH recorded at a time point just before **complete neutralization** takes place.

Point 3: This is the **equivalence point** (halfway up the steep curve). At this point, **moles of NaOH added = moles of CH_3COOH** in the analyte. The H_3O^+ ions are completely neutralized by OH^- ions. The solution contains only CH_3COONa salt and H_2O

Point 4: Beyond the equivalence point (when NaOH titration curve



Acid-Base Titration...

- For this example, let's consider the titration of 50.0 mL of 0.100 M acetic acid, CH_3COOH , with 0.200 M NaOH ($K_a = 1.75 \times 10^{-5}$).
- **Step 1:** We start by calculating the volume of NaOH needed to reach the equivalence point; thus,

$$\text{moles CH}_3\text{COOH} = \text{moles NaOH}$$

$$- M_a \times V_a = M_b \times V_b$$

$$- V_b = \frac{M_a \times V_a}{M_b} = \frac{(0.100 \text{ M})(50.0 \text{ mL})}{0.200 \text{ M}} = 25.00 \text{ mL}$$

- **Step 2: At the beginning of the titration**
- Before adding NaOH the pH is that for a solution of 0.100M acetic acid. Because acetic acid is a weak acid, we calculate the pH using:

Acid-Base Titration...

	$\text{CH}_3\text{COOH}(\text{aq}) + \text{H}_2\text{O}(\ell) \rightleftharpoons \text{CH}_3\text{COO}^-(\text{aq}) + \text{H}_3\text{O}^+(\text{aq})$			
I (Initial)	0.100	–	0	0
C (Change)	–x	–	+x	+x
E (Equilibrium)	$0.100 - x$	–	x	x



$$K_a = \frac{[\text{H}_3\text{O}^+][\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = \frac{(X)(X)}{0.100 - X} = 1.75 \times 10^{-5}$$

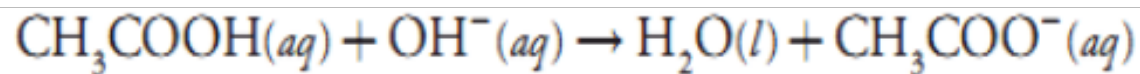
$$X = [\text{H}_3\text{O}^+] = 1.32 \times 10^{-3} \text{ M}$$

- At the beginning of the titration the pH is 2.88

Acid-Base Titration...

Step 3: Before the equivalent point

- Adding NaOH converts a portion of the acetic acid to its conjugate base, CH_3COO^-



- Any solution containing comparable amounts of a weak acid, **HA**, and its conjugate weak base, **A⁻**, is a buffer.
- We can calculate the pH of a buffer using the Henderson–Hasselbalch equation

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

Step 3: Before the equivalent point...

Before the equivalence point the concentration of unreacted acetic acid is

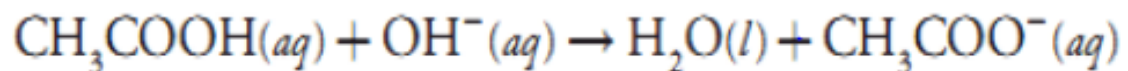
$$\begin{aligned} [\text{CH}_3\text{COOH}] &= \frac{\text{initial moles CH}_3\text{COOH} - \text{moles NaOH added}}{\text{total volume}} \\ &= \frac{M_a V_a - M_b V_b}{V_a + V_b} \end{aligned}$$

and the concentration of acetate is

$$[\text{CH}_3\text{COO}^-] = \frac{\text{moles NaOH added}}{\text{total volume}} = \frac{M_b V_b}{V_a + V_b}$$

Acid-Base Titration...

- **Step 3: Before the equivalent point...**
- For example, after adding 10.0 mL of NaOH the concentrations of CH_3COOH and CH_3COO^- are



- $[\text{CH}_3\text{COOH}] = \frac{(0.100 \text{ M})(50.00 \text{ mL}) - (0.2 \text{ M})(10 \text{ mL})}{50 \text{ mL} + 10 \text{ mL}} = 0.05 \text{ M}$
- $[\text{CH}_3\text{COO}^-] = \frac{(0.2 \text{ M})(10 \text{ mL})}{50 \text{ mL} + 10 \text{ mL}} = 0.0333 \text{ M}$, which gives us a pH of
- $\text{pH} = \text{pK}_a + \log \frac{[\text{CH}_3\text{COO}^-]}{[\text{CH}_3\text{COOH}]} = 4.76 + \log \frac{0.0333 \text{ M}}{0.05 \text{ M}} = 4.58$

Acid-Base Titration...

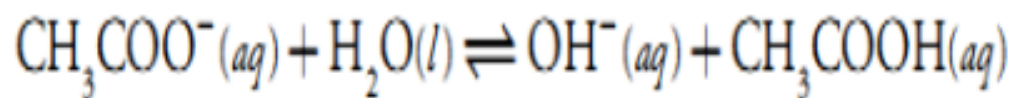
Step 4: At the equivalence point

- The moles of **CH₃COOH** initially present and the **moles of NaOH** added are identical.
- Because their reaction effectively proceeds to completion, the predominate ion in solution is **CH₃COO⁻**, which is a **weak base**.
- To calculate the pH we first determine the concentration of CH₃COO⁻

$$\begin{aligned} [\text{CH}_3\text{COO}^-] &= \frac{\text{moles NaOH added}}{\text{total volume}} \\ &= \frac{(0.200 \text{ M})(25.0 \text{ mL})}{50.0 \text{ mL} + 25.0 \text{ mL}} = 0.0667 \text{ M} \end{aligned}$$

Acid-Base Titration...

- Next, we calculate the pH of the weak base



$$K_b = \frac{[\text{OH}^-][\text{CH}_3\text{COOH}]}{[\text{CH}_3\text{COO}^-]} = \frac{(x)(x)}{0.0667 - x} = 5.71 \times 10^{-10}$$

$$x = [\text{OH}^-] = 6.17 \times 10^{-6} \text{ M}$$

$$[\text{H}_3\text{O}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1.00 \times 10^{-14}}{6.17 \times 10^{-6}} = 1.62 \times 10^{-9} \text{ M}$$

The pH at the equivalence point is 8.79.

Acid-Base Titration...

Step 5: Calculate pH values **after the equivalence point** by determining the concentration of excess titrant.

- For example, after adding 30.0 mL of NaOH, the concentration of OH⁻ is

$$[\text{OH}^-] = \frac{(0.200 \text{ M})(30.0 \text{ mL}) - (0.100 \text{ M})(50.0 \text{ mL})}{50.0 \text{ mL} + 30.0 \text{ mL}} = 0.0125 \text{ M}$$

$$[\text{H}_3\text{O}^+] = \frac{K_w}{[\text{OH}^-]} = \frac{1.00 \times 10^{-14}}{0.0125 \text{ M}} = 8.00 \times 10^{-13} \text{ M}$$

$$\text{pH} = 12.10$$

Acid-Base Titration...

III. Direct titration of weak bases with standard strong acid

- The pH of the solution at the equivalence point, $\text{pH} < 7$
- **Methyl red** is the indicator used most frequently
- **Before equivalence point**, a buffer is formed and the pH does not change very much with added acid
- **After the equivalence point**, pH depends on amount of excess acid added

Practice Exercise

- Construct a titration curve for the titration of 25.0 mL of 0.125 M NH_3 with 0.0625 M HCl (K_b of NH_3 is 1.75×10^{-5})

Acid-Base Titration...

IV. Direct titration of strong base with strong acid

- The pH change in the region of the equivalence point are sufficiently large
- **Methyl orange** is usually used as indicator (4-10).
- **Before the equivalence point**, pH depends on amount of excess strong base present
- **After equivalence point**, pH depends on amount of excess acid added

Practice Exercise

- Construct a titration curve for the titration of 25.0 mL of 0.125 M NaOH with 0.0625 M HCl.

Acid-Base Titration...

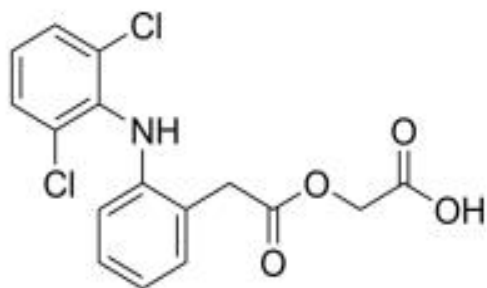
Application of Acid-Base Titration

- Water acidity (CO_2 , S, $\text{Fe}(\text{SO}_4)_3$, FeCl_3)
- Water alkalinity (Hydroxides, carbonates & bicarbonates)
- Water hardness (Ca^{2+} , Mg^{2+} , Fe^{2+} , Mn^{2+})
- Determination of free fatty acids in oils and fats
- Analysis of drugs
 - Acidic drugs
 - Basic drugs

Determination of some pharmaceutically important acids

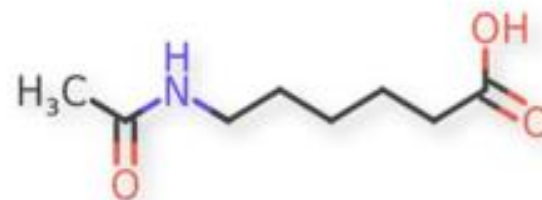
a) Carboxylic acid derivatives;

Aceclofenac (analgesic-anti-inflammatory)



Acexamic acid

Used for treatment of peptic ulcers



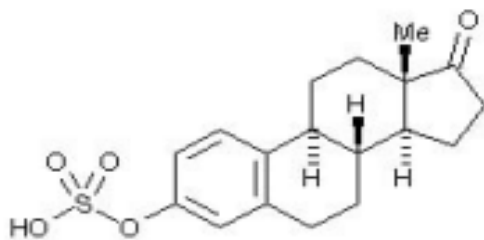
b) Carboxylic acid and phenolic derivatives

Aspirin, Antipyretic

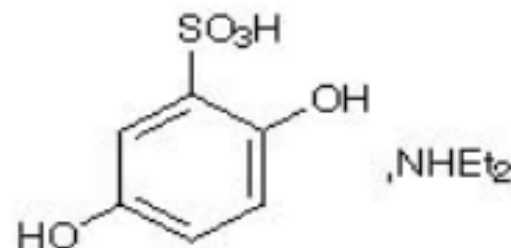
Levodopa, Used for treatment of Parkinson's disease

Determination of some pharmaceutically important acids

c) Sulphonic acid derivatives:

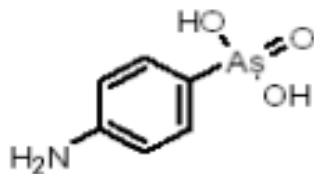


Estropipatic acid

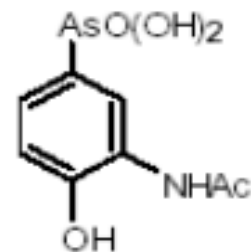


Etamsylate, Haemostatic agent

d) Arsenic acid derivatives;



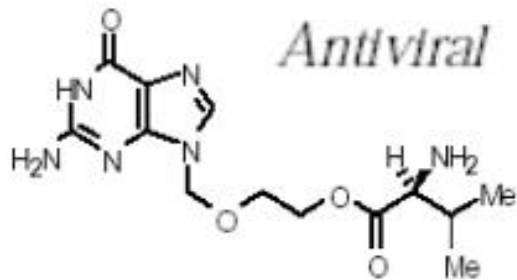
Arsenillic acid, Antibacterial



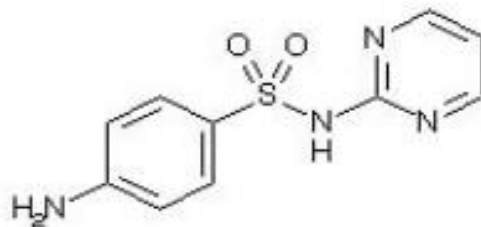
Acetarsol

Determination of some pharmaceutically important bases:

a) Primary amine compounds:

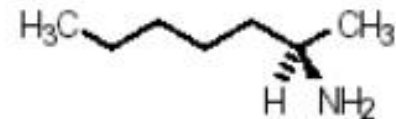


Valaciclovir va·la·sī·klō·veer

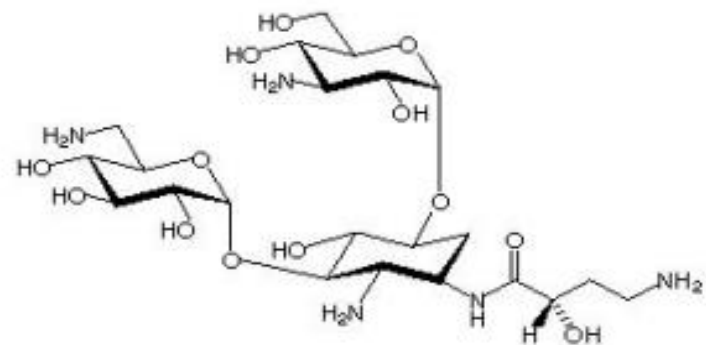


Sulphadiazine

✧ Tuaminoheptane



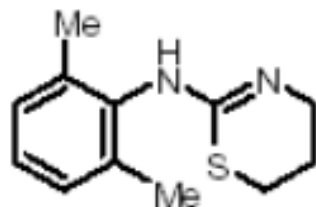
Amikacin, antibiotic



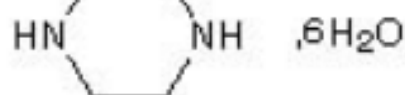
Tedman B. (2013)

b) Secondary amine compounds:

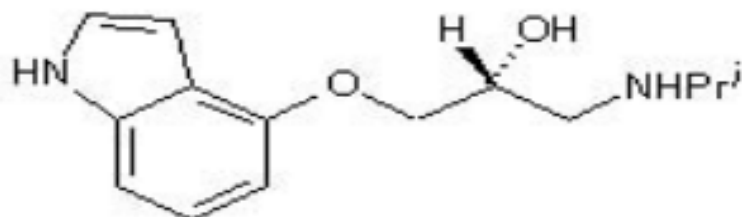
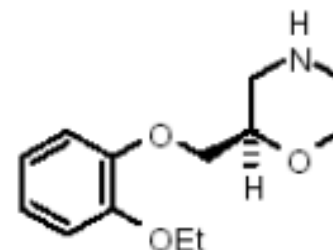
Xylazine, Analgesic



Piprazine hydrate

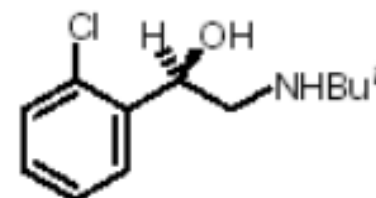


Viloxazine, Antidepressant



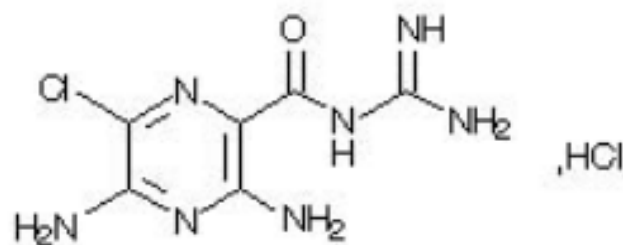
and enantiomer

Pindolol: Beta-adrenoceptor antagonist



Tulobuterol: (Beta-adrenoceptor agonist)

C- Acid Salts of Basic Drugs



Amiloride hydrochloride

Precipitation Titration

Learning Objectives

At the end of this lesson, the student will be able to:

- Understand the fundamentals of precipitation titration
- Distinguish the three methods of Precipitation Titration (Mohr's, Volhard's and Fajan's method)
- Realize the advantages and disadvantages of precipitation titration
- Derive the precipitation titration curve

Precipitation Titration

- It is a type of titration which involves the **formation of precipitate** during the titration technique.
- In precipitation titration, the **titrant reacts** with the **analyte** and forms an **insoluble substance** called a **precipitate**.
 - The titration is continued till the last drop of the analyte is consumed.
- When the **titrant is excess**, it reacts with the **indicator**
 - This is a **signal to terminate the titration process**.

Precipitation Titration

❑ Precipitation titrations are not so popular in present-day routine analysis. Why???

❑ Limitations

- Precipitation reactions are generally:
 - Slow rate of formation of precipitate
 - Involving periods of digestion, cooling, filtration etc., which limit the reactions that are available for titration
 - Difficult to select a **suitable indicator**
 - Difficult to obtain an **accurate precipitate composition**, the **co-precipitation** is a challenge

Precipitation Titration...

☐ Solubility product

- o Solubility is a physical property referring to the **ability for a given substance, the solute, to dissolve in a solvent**
- o It is measured in terms of the maximum amount of solute dissolved in a solvent at equilibrium
 - ✓ **Saturation**
- o Solubility is commonly expressed as a **concentration**
- o An **oversaturated solution** becomes a saturated solution by **forming a solid** to reduce the dissolved material
 - ✓ **Precipitate**

Precipitation Titration...

☐ Solubility product...

- o A saturated solution of a slightly soluble salt in contact with an undissolved salt involves equilibrium like the one below:



- o So, the equilibrium constant expression for the equilibria above is:

$$K_{\text{sp}} = [\text{Ag}^+][\text{Cl}^-]$$

- o This equilibrium constant ***K_{sp}*** is called a **solubility-product constant**

Precipitation Titration...

□ Example-1

- o What is the concentration of silver chloride ions in a saturated silver chloride solution at 25 degrees? $K_{sp} = 1.8 \times 10^{-10}$

✓ Answer: $1.34 \times 10^{-5} \text{ M (mol/L)}$

□ Example-2

- o Calculate the solubility in g/L of calcium sulfate, CaSO_4 , given that its K_{sp} is 2.4×10^{-5} .



✓ Answer: $6.67 \times 10^{-1} \text{ g/L}$

Precipitation Titration...

☐ Factors Affecting Solubility of a Precipitate

1. *Temperature effect*

- ✓ Most of inorganic salts increase its solubility by increasing temperature.
- ✓ Exception: Hg_2Cl_2 (mineral calomel), MgNH_4PO_4 (Struvite)

2. *Solvent effect*

- ✓ Most of inorganic salts dissolve in water but not in organic solvent.
- ✓ **In order to form a precipitate, the product formed should be insoluble in the solvent.**

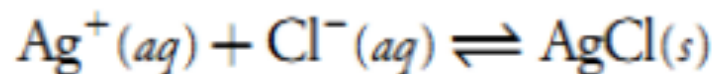
3. *The Common Ion Effect*

- ✓ Precipitate dissolves more easily in water than in solution containing common ions.
- ✓ **Consider:** Solution of AgCl in the presence of NaCl

Precipitation Titration...

Calculating the Titration Curve

- Let's calculate the titration curve for the titration of 50.0 mL of 0.0500 M NaCl with 0.100 M AgNO₃.
The reaction in this case is



- K_{sp} = 1.8 x 10⁻¹⁰, we assume that Ag⁺ & Cl⁻ react completely

Precipitation Titration...

- **Step 1:** Calculate the volume of AgNO_3 needed to reach the equivalence point.

– Moles of Ag^+ = Moles of Cl^-

$$- M_{\text{Ag}} \times V_{\text{Ag}} = M_{\text{Cl}} \times V_{\text{Cl}}$$

Solving for the volume of Ag^+

$$- V_{\text{Ag}} = V_{\text{eq}} = \frac{M_{\text{Cl}} V_{\text{Cl}}}{M_{\text{Ag}}} = \frac{(0.05\text{M})(50\text{mL})}{(0.1\text{M})} = 25 \text{ mL}$$

Precipitation Titration...

- **Step 2:** Calculate pCl before the equivalence point by determining the concentration of unreacted NaCl.
- Before the equivalence point the titrand, Cl^- , is in excess. The concentration of unreacted Cl^- after adding 10.0 mL of Ag^+ , for example, is

$$[\text{Cl}^-] = \frac{\text{initial moles } \text{Cl}^- - \text{moles } \text{Ag}^+ \text{ added}}{\text{total volume}} = \frac{M_{\text{Cl}} V_{\text{Cl}} - M_{\text{Ag}} V_{\text{Ag}}}{V_{\text{Cl}} + V_{\text{Ag}}}$$

$$[\text{Cl}^-] = \frac{(0.0500 \text{ M})(50.0 \text{ mL}) - (0.100 \text{ M})(10.0 \text{ mL})}{50.0 \text{ mL} + 10.0 \text{ mL}} = 2.50 \times 10^{-2} \text{ M}$$

$$\text{pCl} = -\log [\text{Cl}^-] = -\log (2.50 \times 10^{-2}) = 1.60$$

Precipitation Titration...

- **Step 3:** Calculate pCl at the equivalence point using the Ksp for AgCl to calculate the concentration of Cl⁻
- At the titration's equivalence point, we know that the concentrations of Ag⁺ and Cl⁻ are equal.
- To calculate the concentration of Cl⁻, we use the Ksp expression for AgCl; thus

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] = (x)(x) = 1.8 \times 10^{-10}$$

Solving for x gives $[\text{Cl}^-]$ as 1.3×10^{-5} M, or a pCl of 4.89

Precipitation Titration...

- **Step 4:** Calculate pCl after the equivalence point by first calculating the concentration of excess AgNO_3 and then calculating the concentration of Cl^- using the K_{sp} for AgCl .
- For example, after adding 35.0 mL of titrant

$$[\text{Ag}^+] = \frac{\text{moles Ag}^+ \text{ added} - \text{initial moles Cl}^-}{\text{total volume}} = \frac{M_{\text{Ag}}V_{\text{Ag}} - M_{\text{Cl}}V_{\text{Cl}}}{V_{\text{Cl}} + V_{\text{Ag}}}$$
$$= \frac{(0.100 \text{ M})(35.0 \text{ mL}) - (0.0500 \text{ M})(50.0 \text{ mL})}{50.0 \text{ mL} + 35.0 \text{ mL}} = 1.18 \times 10^{-2} \text{ M}$$

$$K_{sp} = [\text{Ag}^+][\text{Cl}^-] \text{ then } [\text{Cl}^-] = \frac{K_{sp}}{[\text{Ag}^+]} = \frac{1.8 \times 10^{-10}}{1.18 \times 10^{-2}} = 1.5 \times 10^{-8} \text{ M}$$

$$\text{pCl} = -\log[\text{Cl}^-] = -\log(1.5 \times 10^{-8}) = 7.8\text{M}$$

Precipitation titration...

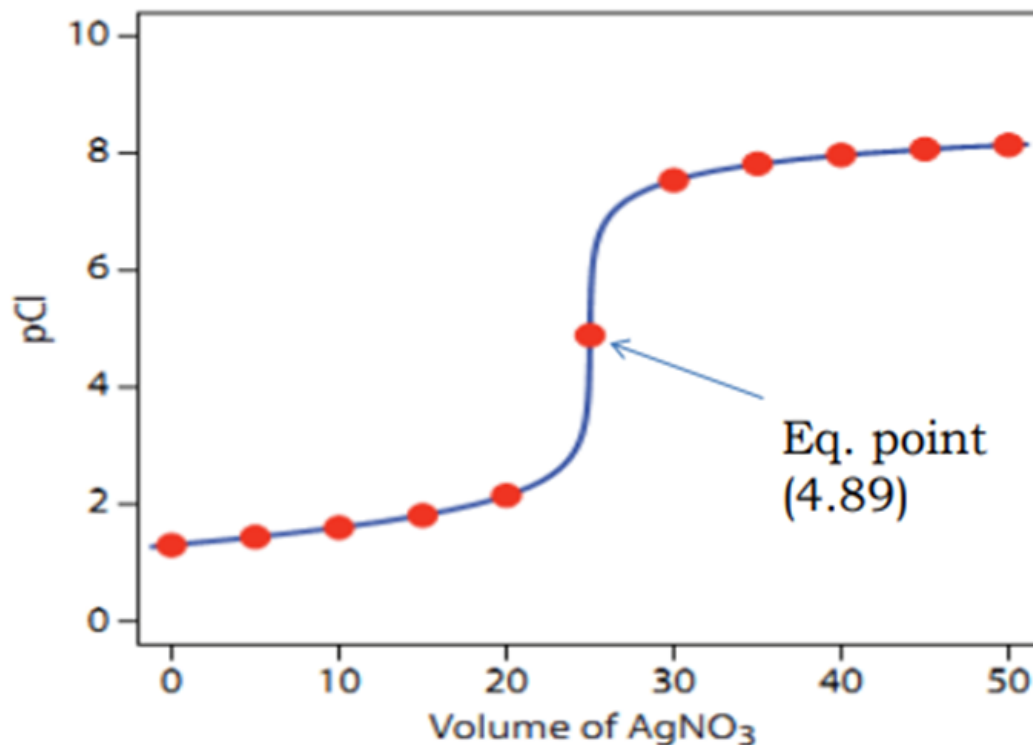


Fig. Titration curve for the titration of 50.0 mL of 0.0500 M NaCl with 0.100 M AgNO_3 .

Precipitation Titration...

Argentometric Titrations:

- Argentometry is derived from a Latin word *Argentum* which means silver.
- The **most widely applicable precipitation titrations** involve the use of **silver nitrate** with **chlorides, bromides, iodides, and thiocyanate**.
- Since silver is always there, **precipitation titrations with silver nitrates** are referred to as **Argentometric titrations**.
- According to **End Point Detection Method**, **three main precipitation methods** are widely used depending on the type of application.
 - These are:
 - **Mohr's Method,**
 - **Volhard's Method and**
 - **Fajan's Method**

Precipitation Titration...



End-point Detection



Mohr's Method

- Chromate ion (CrO_4^{2-}) is added to serve as indicator.
- Mohr titration has to be performed at a **neutral or weak basic solution of pH 7-9. WHY?**
- **At low pH (<6)- over titration**



- **In higher pH (basic), Ag_2O will be formed** as a black precipitate and hence, silver ion will be over consumed there by resulting in titration error.
- After equivalent point, **silver chromate (red color)** is formed

Precipitation Titration...

➡ End-point Detection...

➤ Mohr's Method...

- Mohr method it is **not feasible** for I^- and SCN^- because of
 - **Adsorption problem**
 - The chromate is adsorbed and hence, **too low conc. of the chromate** will be available to **react with excess silver ion**; end point detection is very difficult
 - Silver iodide is **bright yellow** in color which **interferes** with the **end point color**.

Precipitation Titration...

➡ End-point Detection...

➤ Mohr's Method...

- The prominent usage of Mohr's method is in **detecting the presence of halides** in water samples.
- Detection of halides in water samples is an **important health safety measure**, and Mohr's method is an imperative need to analyse different water samples to check **how safe they are**.

Precipitation Titration...

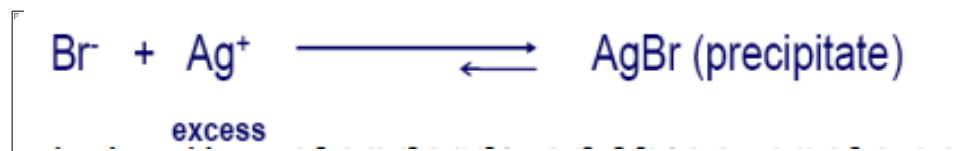


End-point Detection...



Volhard's Method

- o Volhard titration is an indirect titration (**back titration**)
- o **Titration principle:** Silver solution is added excessively to a halide.



- o Back titrated with a **standardized thiocyanate solution**.
- o **Fe(III) indicator** is added



- o **Reddish brown ferric thiocyanate** is formed at the end point
- o This is **Sensitive and very precise** method
- o It is used to detect the concentration of **halides** in a given component.
- o It is also key in detecting anions along with **potassium and fluoride**.

Precipitation Titration...

Advantages of Volhard's method:

- The **acidic environment** give advantage for **halide analysis** because anions such as **carbonate**, **oxalate** and **arsenate** that do not form precipitate with silver in acidic medium (but they do in basic medium) will **not interfere with halides**
- Give **accurate results** due to back titration

Limitations of Volhard's method:

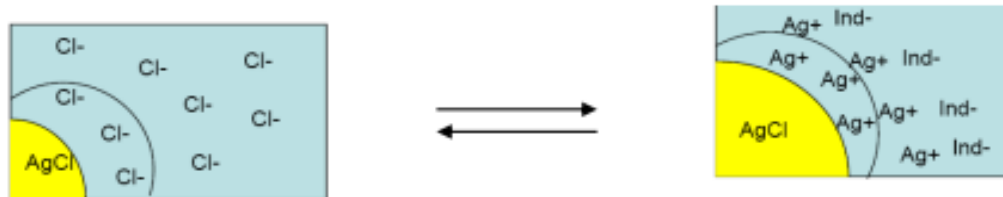
- Can **not be used in neutral or basic medium**
- **Time consuming**

Precipitation Titration...

➡ End-point Detection...

➤ Fajan's Method

- The reaction between the indicator and excess silver takes place on the surface of the precipitate



- The color of the **adsorbed indicator** is different from that of **unadsorbed indicator** and this difference signals the completion of the titration

Example: Fluorescein (at neutral or basic pH)

Summary

Mohr vs Volhard vs Fajans Method

Mohr Method

Mohr method is an analytical technique in which we can determine the halide concentration through direct titration

Volhard Method

Volhard method is an analytical technique in which we can determine the halide concentration through a back titration

Fajans Method

Fajans method is an analytical technique in which we can determine the halide concentration through adsorption

TECHNIQUE

Direct titration

Back titration

Adsorption followed by back titration

INDICATOR

Chromate

Ferric ions

Fluorescein

METHOD

Directly titrating the chloride ions using the reaction between silver ions and chloride ions

Indirect titration of chloride ions with an excess amount of silver ions, followed by the determination of the excess silver ion content through titration with thiocyanate ions

Adsorption of fluorescein onto the surface of silver chloride colloidal surface followed by determination of end point using the reaction between silver ion and fluorescein

Complexometric Titration

Learning Objectives

At the end of this lesson, the student will be able to:

- Explain Complexometric Titration
- Discuss the Principle of Complexometric Titration
- Describe the Classification of Ligands
- Discuss the Factors that determine the magnitude of break in titration curve at end point
- Compare the Types of Complexometric Titrations
- Describe the Applications of Complexometric Titrations

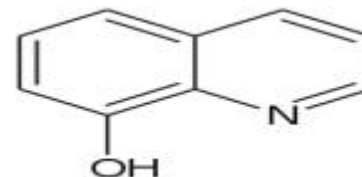
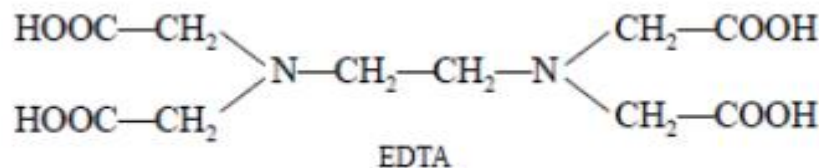
Complexometric Titration (sometimes chelatometry)

- It is a form of volumetric analysis in which the formation of a **colored complex** is used to **indicate the end point of a titration**.
- Particularly useful for the **determination of a mixture of different metal ions in solution**.
- In the reaction, a simple ion is transformed into a complex ion and the equivalence point is determined by using **metal indicators or electrometrically**.
 - It is a method used in **quantitative chemical analysis**.
- In this type of titration, a sample is titrated with a **complexing agent** in order to determine the amount of a given species present in the sample.
 - A complexing agent is a substance which forms a complex with the analyte
- A **change in colour of an indicator** is used to detect when the **end-point has been reached**
 - An indicator capable of producing an unambiguous color change is usually used to detect the end-point of the titration.

Complexometric Titration...

Chelating Agent

- An organic agent that has two or more groups capable of complexing with a metal ion is called a **chelating agent**
 - It has **more than one electron donating groups**
- The **complex formed** is called a **chelate**.
- The **chelating agent** is called the **ligand**.
- The most effective complexing agent in ligands are **amino** and **carboxylate ions**.
- Titration with a chelating agent is called a **chelometric titration**, perhaps the most important and practical type of complexometric titrations.
 - For example **Ethylenediaminetetraacetic acid (EDTA)** and **8-hydroxy quinoline** are important reagents used.



Complexometric Titration ...

- **Classification of Ligands**

- **Unidentate Ligands (Monodentate Ligand)**
- **Multidentate Ligands**
 - **Bidentate ligands** (2 donor atoms)
 - **Tridentate ligands** (3 donor atoms),
 - **Hexadentate ligands** (6 donor atoms)

- **Unidentate Ligands (Monodentate Ligand):**

- Ligands that are bound to metal ion **only at one place**
- Ligand that contain only one atom that can be attached to central metal.
- **Halide ions, cyanide ions, and NH_3** are common examples of unidentate ligands.
 - ✓ NH_3 , for example, is a unidentate ligand capable of complexing with **cupric ions**.

Formation of Complex with Monodentate Ligand



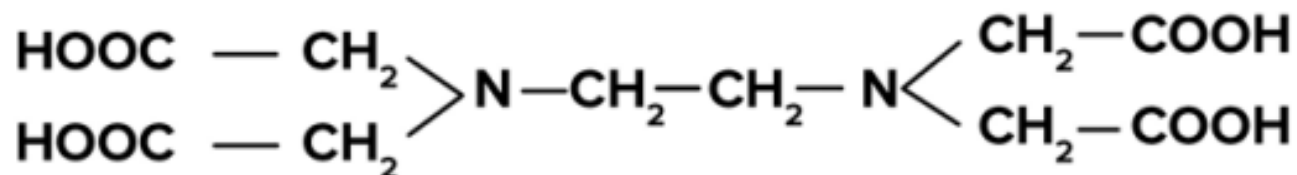
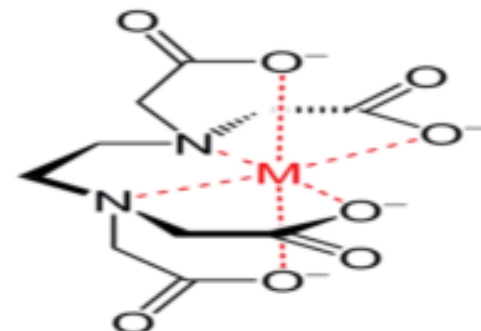
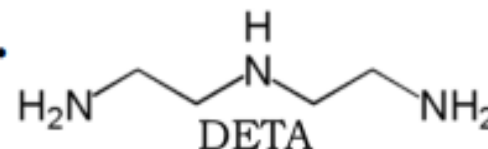
Considering the overall reaction:



Complexometric Titration ...

Multidentate Ligands

- Many ligands are known that contain more than one group, capable of binding with metal ions.
 - They are called **multidentate ligands**.
- They include:
 - Bidentate ligands** (2 donor atoms)
 - E.g. Ethylenediamine $\text{H}_2\text{N}-\text{CH}_2-\text{CH}_2-\text{NH}_2$
 - Tridentate ligands** (3 donor atoms),
 - E.g. **Diethylenetriamine (DETA)**
 - Hexadentate ligands** (6 donor atoms)
- EDTA is most commonly available as the disodium salt, where the two ionized carboxylic acid groups form the salt



Complexometric Titration ...

Ethylenediaminetetraacetic acid (EDTA)

- It forms chelates with **nearly all metal ions**
- Such titrations are **called complexometric or chelometric or EDTA titrations.**
- Disodium salt of EDTA is a water soluble chelating agent and is always preferred.
- It is **non-hygroscopic and a very stable sequestering agent**
 - Ligands which form water soluble chelates are called sequestering agents
- It forms complexes with most cations **in a 1:1 ratio**, irrespective of the valency of the ion.

Complexometric Titration...

- **Stability of Complexes** : The general equation for the formation of a 1:1 chelate complex, MX, is:



- **Stability Constant (K)** = $\frac{[MX]}{[M][X]}$

- **NOTE:** Increase in temperature causes a slight **increase in the ionization** of the complex & a **slight lowering of K**.
- **The presence of electrolytes having no ion in common with the complex decreases K**, whilst the **presence of ethanol increases K**, probably due to the suppression of ionization.

Complexometric Titration...

Principle of Complexometric Titration

- In complexometric titration:
 - The free metal ions disappear as they are changed into complex ions.
 - End point can be detected usually **with an indicator** or instrumentally by **potentiometric or conductometric (electrometric)** method.

Complexometric Titration...

End Point Visualization Method:

- In a complexometric titration, the detection of the endpoint can be observed by the two methods.
 - Visual &
 - Instrumental Methods

Visual Method

- It is the **most common method** because it is **accurate** and **cost efficient**.
- Examples of the visual methods used for determining the endpoint of the complexometric titrations:
 - Metallochromic indicators, pH indicators, or Redox indicators

Instrumental Method

- There is the possibility of human error and inaccuracy in visual methods in determining the endpoint.
- Therefore, some instrumental techniques are used in endpoint determination.
- The examples of instrumentation methods are as follows:
 - **Potentiometry**, Photometry, or Miscellaneous methods.

Complexometric Titration...

Methods of End Point Detection: Criteria for the indicators

- Metal indicators must comply with the following requirements:
 - Compound must be **chemically stable** throughout the titration.
 - It should form **1:1 complex** which must be **weaker than the metal chelate complex**.
 - Colour of the indicator and the metal complexed indicator must be **sufficiently different**.
 - The indicator **should not compete with the EDTA**.

Complexometric Titration ...

Factors that determine the magnitude of break in titration curve at end point

- **There are three factors** that are important in determining the **magnitude** of break (rapid changes) in titration curve at end point.
 - **The stability of complex formed**
 - **The number of steps involved in complex formation &**
 - **Effect of pH**
- **1) The stability of complex formed:**
 - **The greater the stability constant (K) for complex formed, the larger the change in free metal concentration (pM) at equivalent point.**
- **2) The number of steps involved in complex formation:**
 - **The Fewer the number of steps** required in the formation of complex, the **greater would be the break in titration** curve at equivalent point.

Complexometric Titration ...

Factors that determine the magnitude of break in titration curve at end point...

• 3) Effect of pH:

- During a complexometric titration, **the pH must be constant by use of a buffer solution.**
- Control of pH is important since **the H⁺ ion plays an important role in chelation.**
- **Most ligands are basic and bind to H⁺ ions** throughout a wide range of pH.
 - Some of these H⁺ ions are frequently displaced from the ligands (chelating agents) by the metal during chelate formation.
- Equation below shows complexation between metal ion and H⁺ ion for ligand:



- Thus, **stability of metal complex is pH dependent**

Complexometric Titration ...

Factors that determine the magnitude of break in titration curve at end point...

- **3) Effect of pH...**

- **The Lower the pH of the solution, the lesser would be the stability of complex** (because more H^+ ions are available to compete with the metal ions for ligand).
- Only metals that form **very stable complexes can be titrated in acidic solution**, and **metals forming weak complexes** can only be effectively titrated in **alkaline solution**.

Complexometric Titration...

Types of Complexometric Titrations

- In Complexometric titrations, there are of 4 types:
 - **Direct Titration, Back Titration, Replacement Titration** and **Indirect Titration**
- **Direct Titration:**
 - It is the **most convenient** and **simplest method** of complexometric titration using EDTA.
 - It is similar to the acid–base titration technique.
 - The standard, EDTA solution is added to give sample containing metals using burette till the endpoint is achieved.
 - Copper, barium, zinc, mercury, aluminium, lead, bismuth, chromium, etc. are metals that can be determined by using direct complexometric titration.
- **Limitations:**
 - Slow complexation reaction
 - Interference due to presence of other ions

Complexometric Titration...

Types of Complexometric Titrations...

- **Back Titration:**
 - In this method, an excess amount of standard solution of EDTA is added to the metal solution being examined.
 - Then the **excess amount of EDTA** is back titrated by the solution of the second metal ion.
 - E.g. **Determination of Mn**. This metal cannot be directly titrated with EDTA because of **precipitation of Mn(OH)_2** .
 - It is back titrated with a **standard Zn solution** kept in burette using **Eriochrome Black T** as indicator.

Complexometric Titration...

Types of Complexometric Titrations...

- **Replacement Titration:**

- It is used when direct titration or back titration **don't give sharp endpoints.**
- This analyte (containing metal) is added to the metal–EDTA complex.
- The metal present in the analyte displaces quantitatively another metal from the metal–EDTA complex.



- The **freed Mg metal** is then directly titrated with a **standard EDTA solution.**
- Mn quantitatively displaces Mg from MgEDTA chelate.
- This displacement takes place because Mn forms a **more stable complex** with EDTA.
- By this method **Ca, Pb, Hg** may be determined using **Eriochrome Black T indicator.**

Complexometric Titration...

Types of Complexometric Titrations...

- **Indirect Titration:**

- It is also known as **Alkalimetric titration**.
- Some **anions** form a **precipitate with metal cations**, which **do not react** with EDTA chelate.
- So, these ions can be analyzed by indirect titration with EDTA
 - Protons from disodium EDTA are displaced by a heavy metal and titrated with sodium alkali.
- **For example: Barium ions can be determined by indirect titration.**
 - Barbiturates also do not react with EDTA but are quantitatively precipitated from **alkaline solution** by **mercuric ions** as 1:1 complex.

Complexometric Titration...

Applications of Complexometric Titrations

- It is widely used in the pharmaceutical industry to **determine the metal conc. in drugs.**
 - Titanium dioxide is used in many cosmetic products, which can be analyzed by complexometric titration.
- **Determination of various metals** like Ca, Mg, Pb, Zn, Al, Fe, Mn, Cr, etc. in different formulations that are official in pharmacopeia
- It is used to **analyze urine samples.**
- It is also used for the **determination of total Hardness in water.**

Redox Titration

Learning Objectives

At the end of this lesson, the student will be able to:

- Explain about Redox Titration
- Describe the Applications of Redox Titrations
- List down oxidimetric and reductimetric reagents
- Describe the Application of diazotization titration
- List some compounds that can be assayed by sodium nitrite titration in diazotization titration

Redox Titration (Oxidation-Reduction Titrations)

- It is a laboratory method of determining the conc. of a given analyte by causing a **redox reaction** b/n the titrant & the analyte
 - These types of titrations sometimes require the use of a **potentiometer or a redox indicator** to show the end-point of the titration
- A common example of a redox titration is treating a solution of **iodine with a reducing agent to produce iodide** using a starch indicator to help detect the endpoint.

- **Iodine (I_2) can be reduced to iodide (I^-) by thiosulfate ($S_2O_3^{2-}$),**

and when **all iodine is spent**, the **blue colour disappears**.

- This is called an **iodometric titration**.
- This type of titration is used to measure the amount of an **oxidizable or reducible species present in the sample**.

Redox Titration...

- **Redox titration:** is titration in which the reaction between the analyte and titrant is an **oxidation/reduction reaction**.
- Titrations can also be based on chemical reactions that **transfer electrons**.
- A redox titration can be described by the following equation:
$$\text{Ox1} + \text{Red2} \rightarrow \text{Red1} + \text{Ox2}$$
- These titrations involve the titration of an **oxidizing agent (or oxidant) with a reducing agent (or Reductant) or vice versa**
- The chemical reaction proceeds **with transfer of electrons** (simultaneous **loss or gain of electrons**) among the reacting ions in aqueous solutions.

Redox Titration...

- **Oxidation** process involves **loss of electrons**
- **Reduction** process involves **gain of electrons**.
- **An oxidizing agent** oxidizes the other substance by stripping off electrons from it.
- **A reducing agent** reduces the other substance by donating electrons to it.



- Here, the **ferrous ions** are getting oxidized to **ferric ions** by the loss of one electron.



- The **chlorine** gets reduced to **chloride ions** by gaining electrons
- Further, oxidation and reduction are **always found to occur together in a reaction.**

in a reaction.

- In other words, **electron loss of one substance is always accompanied by an electron gain** on the part of the other and consequently there are **no free electrons in a chemical system.**

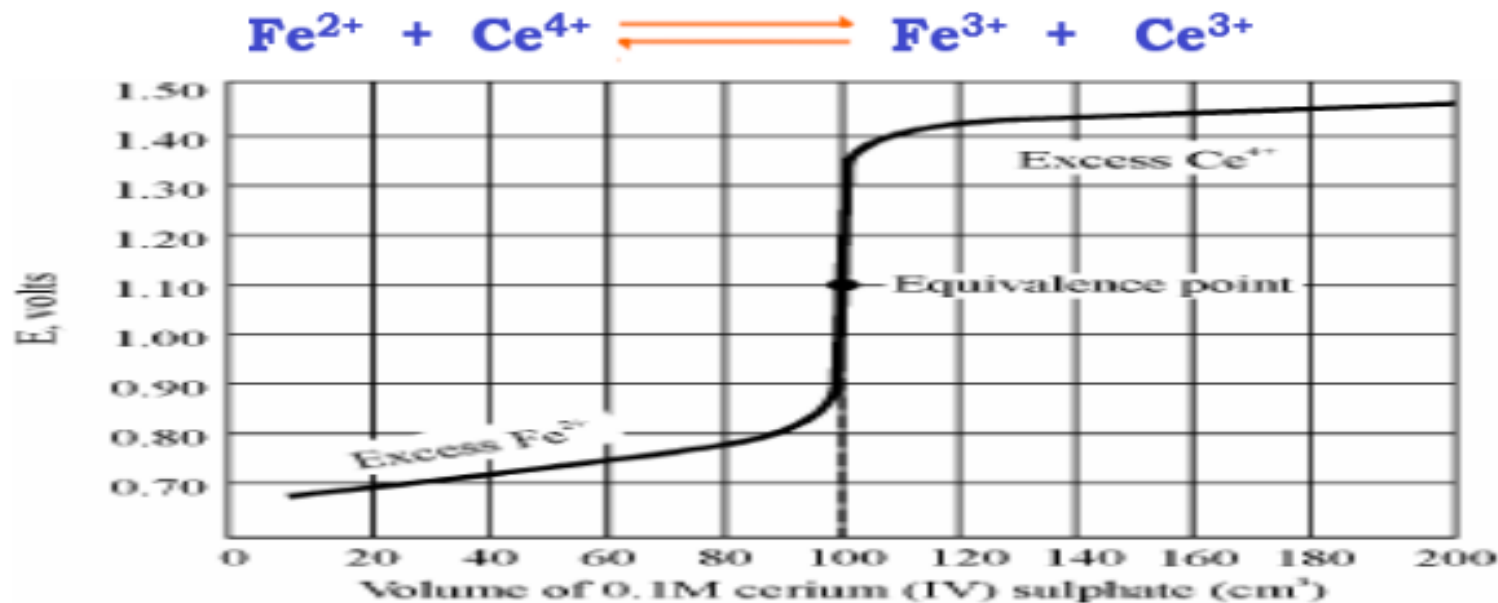
Redox Titration...

- The **relative tendency to accept or lose electrons** by any reagent is measured in terms of their **standard reduction potential values**.
 - **Reduction potential**: is the measure of how favorable it is for a compound to gain electrons.
 - A **high positive value** for reduction potential indicates that a compound is readily reduced and hence, is a **strong oxidizing agent**

Redox Titration...

Redox Titration Curves

- By plotting the **redox potential** corresponding to different points in **the titration**, a titration curve similar to the curve obtained in an acid-base method is obtained.
- Example:** Titration of 100 mL of **0.1M iron (II)** with **0.1 M cerium (IV)** in the presence of dilute sulphuric acid.



Redox Titration...

Redox Indicators

- A redox indicator is a compound which can **undergo a redox reaction** and **change colour depending on the potential of the solution.**
- The oxidized and reduced forms of the indicator **have different colours.**



- The redox indicator will show colour X if the intensity of this colour (oxidized form) is **at least 10 times greater than the intensity of the colour Y.**
- Similarly, the indicator will show **colour Y** if the **intensity of the colour is 10 times more than that of the colour X.**

Redox Titration: Redox Indicators

Table: The colour of the oxidized and reduced form of some redox indicators

Indicator	Color of Oxidized form	Color of Reduced form
Sodium 2,6-Dibromophenol-indophenol or Sodium 2,6-Dichlorophenol-indophenol	blue	colorless
Sodium o-Cresol indophenol	blue	colorless
Thionine (syn. Lauth's violet)	violet	colorless
Methylene blue	blue	colorless
Indigotetrasulfonic acid	blue	colorless
Indigotrisulfonic acid	blue	colorless
Indigo carmine (syn. Indigodisulfonic acid)	blue	colorless
Indigomono sulfonic acid	blue	colorless
Phenosafranin	red	colorless
Safranin T	red-violet	colorless
Neutral red	red	colorless

Redox Titration...

Redox Titration...



Two types of reagents are there: **Oxidimetric Reagents** and **Reductimetric Reagents**

A. Oxidimetric Reagents

- **i) Permanganate Titrations (KMnO_4):** These are titrations in which **potassium permanganate is used as an oxidizing agent**.
 - It is an oxidizing agent in all medium but **it is strong in acidic medium**
 - The medium is **maintained by the use of dilute sulfuric acid**
 - **KMnO_4 acts as a self-indicator (colorless to pink)**
 - The potential equation, when potassium permanganate acts as an oxidizing agent, is:



Redox Titration...



A. Oxidimetric Reagents...

- **ii) Potassium Dichromate Titrations ($K_2Cr_2O_7$):** It is also a very strong oxidizing agent



- However, it is not as strong oxidizing agent as permanganate
- Still it is **widely used in redox titrations** because of several **advantages over permanganate**.
 - Unlike $KMnO_4$, $K_2Cr_2O_7$ is available in high purity
 - The aqueous solutions are **quite stable towards light**.
 - It is an **excellent primary standard** & its standard solutions can be prepared by **direct weighing**
 - But $K_2Cr_2O_7$ acts as oxidizing agent in acidic medium only to determine most compounds that can be titrated with $KMnO_4$
- The end point in titrations with **dichromate ion** is generally detected by using suitable **redox indicators such as diphenylamine, sodium diphenylaminesulphonate and N-phenylanthranilic acid, etc.**

Redox Titration..



A. Oxidimetric Reagents...

- **iii. Cerium (IV) titrations [Cerium (IV) sulphate, $\text{Ce}(\text{SO}_4)_2$]**
 - It is a **powerful oxidizing agent** in **acidic medium**.
 - It can be used **only in acid solution**.
 - As the solution is **neutralized, Cerium(IV) hydroxide or basic salts precipitate**.
- **Advantages**
 - Stable over prolonged periods.
 - They need not be protected from light.
 - It is not too highly colored to obstruct vision when reading the meniscus in burettes

Redox Titration...



A. Oxidimetric Reagents...

• iv. Potassium Bromate Titrations (KBrO₃):

- It is also a **strong oxidant in acidic medium**.
- It is a **primary standard** as the reagent can be obtained in a pure state.
- Its standard solution can be prepared by direct weighing.
- **Yellowish color** due to bromine released from the **slight excess of reagent added...end point**
- The end point may also be detected by indicators such as **methyl orange, methyl red etc.**
- The reagent is used in **hydrochloric acid** for the direct determination of **arsenic(III), antimony(III), tin(II), hydrazine, etc.**



Redox Titration...

A. Oxidimetric Reagents...

• v. Iodometric Titrations (KI):

- An oxidation agent is allowed to react in **neutral medium** or **acidic medium**, with excess of **potassium iodide** to liberate **free iodine**.



- Free iodine is titrated against a standard reducing agent usually with **sodium thiosulphate**.



- The end point can be detected by adding **starch**: **The blue color disappears at the end point**.
- **Halogen, dichromates, cupric ion, peroxides, etc.**, can be estimate by this method.



- **Pharmacopoeia**: **Phenol, Methyl hydroxybenzoate, Phenindione**
- In general,
 - The method in which iodine is used for the determination of **reductants** is called **iodimetry**.
 - The method in which iodide is used for the determination of **oxidants** is called **iodometry**

Redox Titration...

B. Reductimetric Reagents

- Standard solutions of **reducing agents** are **not used as widely as oxidizing agents**, because **most of them are oxidized by dissolved and atmospheric oxygen**.
- They are, therefore, **less convenient to prepare and use**.
- **Thiosulphate ($\text{S}_2\text{O}_3^{2-}$)**: is the **only common reducing agent** that is **stable to air oxidation** and can be kept for **long periods of time**.
- **Other common reductimetric reagents**
 - **Iron**
 - **Chromium (II) and Titanium(III)**

Diazotization Titration or Nitrite Titration

- It is the conversion of any **primary aromatic amine to a diazonium compound**.
 - This process was first discovered in 1853 and was **applied to the synthetic dye industry**.
- The reaction mechanism was first proposed by Peter Griessin.
- In this method, the **primary aromatic amine** is reacted with the **sodium nitrite in acidic medium** to form a **diazonium salt**.
 - This method is first used in the determination of dyes.
- $\text{R-NH}_2 + \text{NaNO}_2 + \text{HCl} \rightarrow \text{R-N}^+ \equiv \text{N-Cl}^- + \text{NaCl} + \text{H}_2\text{O}$



Diazotization Titration...

- Many **aromatic primary amine** with free -NH_2 group can be analyzed quantitatively by measuring the volume of NaNO_2 solution required to convert them into diazonium salts.
- **Observation of end point depends** upon the detection of small **excess of nitrous acid** which in turn can be detected by the **starch paper or paste as an external indicator**.
 - Starch iodide paper or paste is the **starch solution plus equal volume of 5% KI solution in H_2O** .
- Starch iodide paste with reaction mixture is as follows:

- Starch iodide paste with reaction mixture is as follows:



- **The liberated I_2 reacts** with starch to give **blue color**.
- The liberated I_2 reacts with starch to give **blue color**.

Diazotization Titration...

Application of Diazotization Titration

- **Analysis of sulphonamides by diazotization** of primary aromatic amino group usually present in this class of drugs.
- Several sulphonamides require the formation of **primary amine prior to diazotization step**.
- List of compounds assayed by sodium nitrite titration:

- ✓ Dapsone
- ✓ Calcium amino salicylate
- ✓ Sodium amino salicylate
- ✓ Sulphacetamide tablet
- ✓ Sulphadoxine

- ✓ Benzocaine
- ✓ Procaine hydrochloride
- ✓ All sulpha drugs

Thank You